

SiamRPN++ Training Pipeline — Dataflow Debug

Full visualisation of every preprocessing step for each active training dataset

Active Datasets in Current Training Run:

1. Anti-UAV 410

200 sequences, 1500 frames each, 640×512 LWIR

↳ Primary IR SOT dataset — drones against sky

2. MSRS

520 sequences, paired IR/visible, 480×640

↳ Multi-spectral road-scene pairs

3. MassMIND

1801 sequences, 640×512 LWIR

↳ Aerial IR tracking — large diverse set

4. DUT-VTUAV

225 sequences, 1920×1080 thermal UAV

↳ IR UAV tracking, 10fps video / 1fps annotations

Key Config:

EXEMPLAR_SIZE=127 SEARCH_SIZE=255 OUTPUT_SIZE=25 ANCHOR_NUM=5 BATCH=32 EPOCHS=500

Dataflow Steps per Sample:

Step 0: Raw full frame + ground-truth annotation box

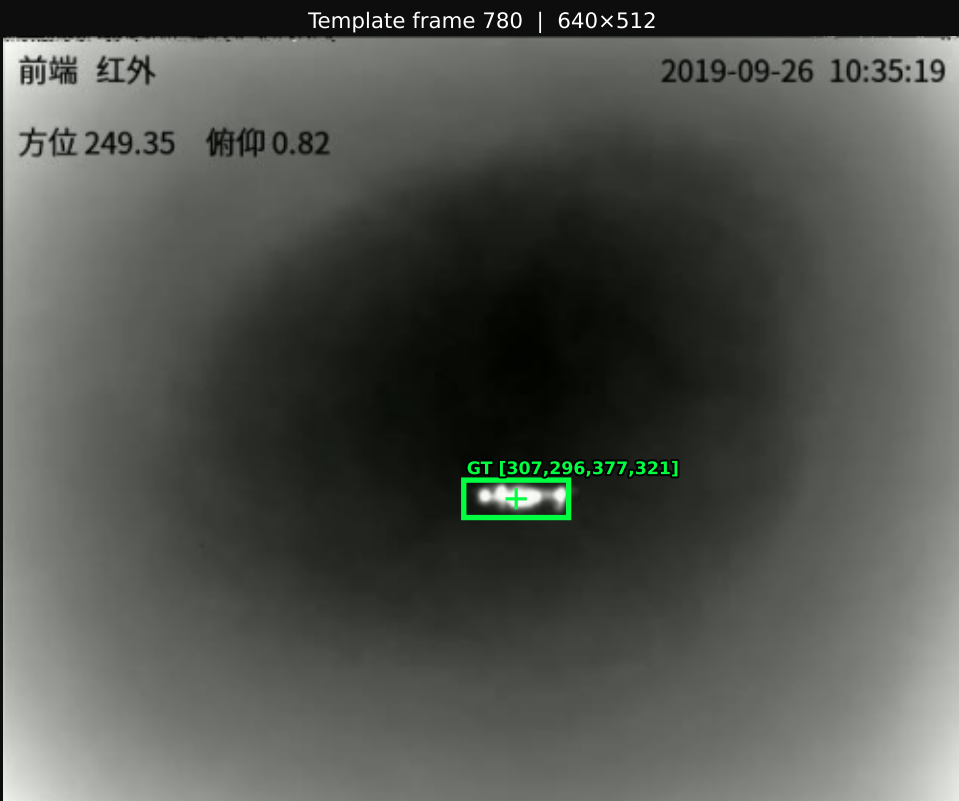
Step 1: `_get_center_crop()` → 254×254 template / 510×510 search

Step 2: `Augmentation()` → 127×127 template / 255×255 search

Step 3: `AnchorTarget()` → cls map (pos/neg/ign) + regression delta

Step 4: Final model tensors (raw 0-255 float32, no normalisation)

STEP 0 – RAW INPUT



```
— TEMPLATE ANNOTATION —
Format : [x1,y1,x2,y2]
Values : [307.0, 296.0, 377.0, 321.0]
Size : 70.0 x 25.0 px
Center : (342.0, 308.5)
AR : 2.89
Image : 640x512

— SEARCH ANNOTATION —
Values : [315.0, 324.0, 384.0, 349.0]
Size : 69.0 x 25.0 px
Center : (349.5, 336.5)
Image : 640x512
```

STEP 1 – CENTER CROP

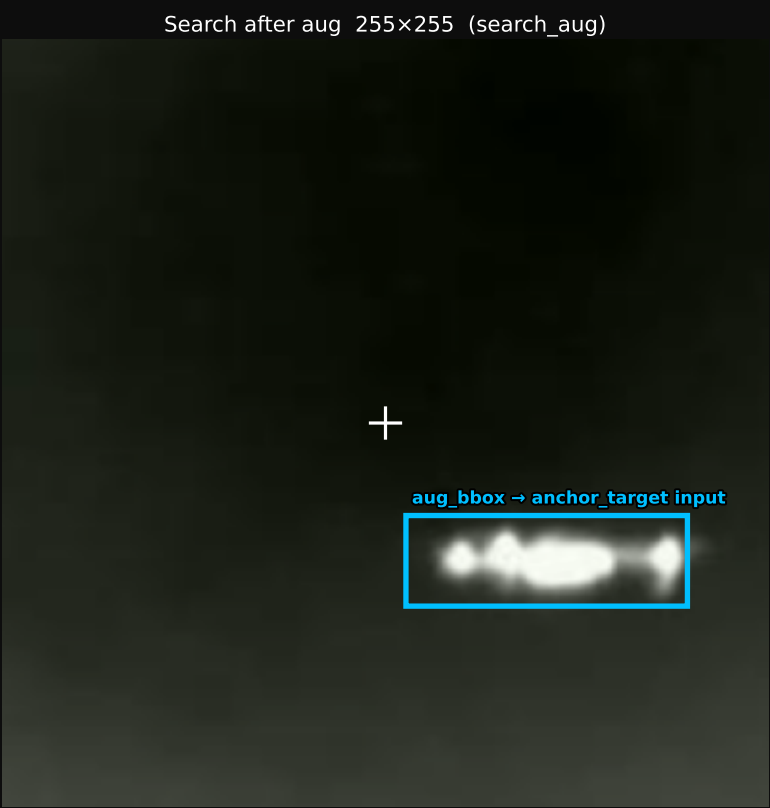
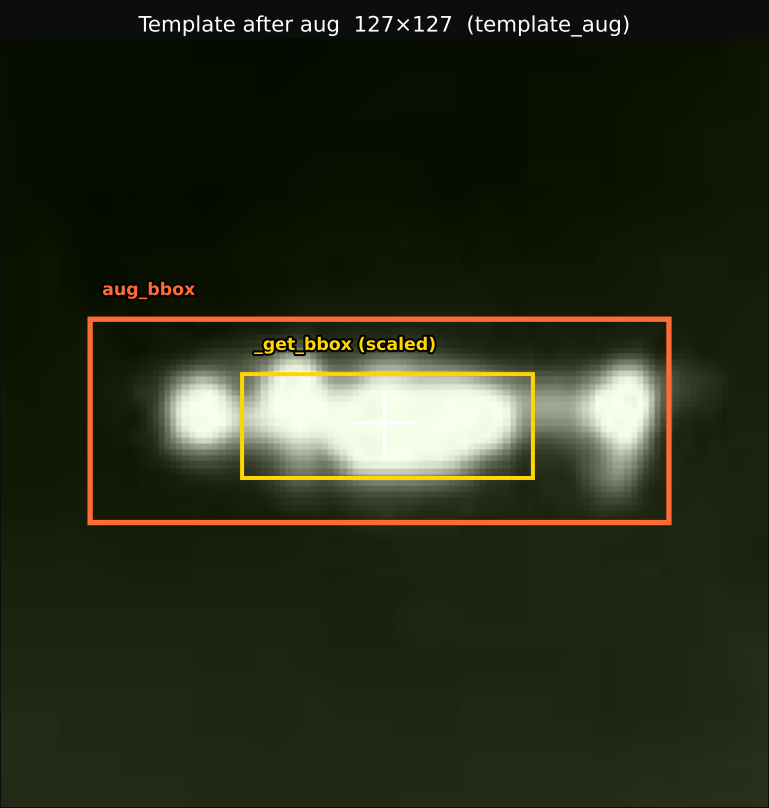


```
— get_center_crop —
Output size (tmpl): 254x254
Output size (srch): 510x510
White cross = image centre
Orange box = new annotation

— get_bbox_output —
TEMPLATE (corner):
[78.8, 109.8, 175.2, 144.2]
cx=127.0 cy=127.0
w=96.3 h=34.4

SEARCH (corner):
[207.1, 237.6, 302.9, 272.4]
cx=255.0 cy=255.0
w=95.9 h=34.7
```

STEP 2 – AUGMENTATION

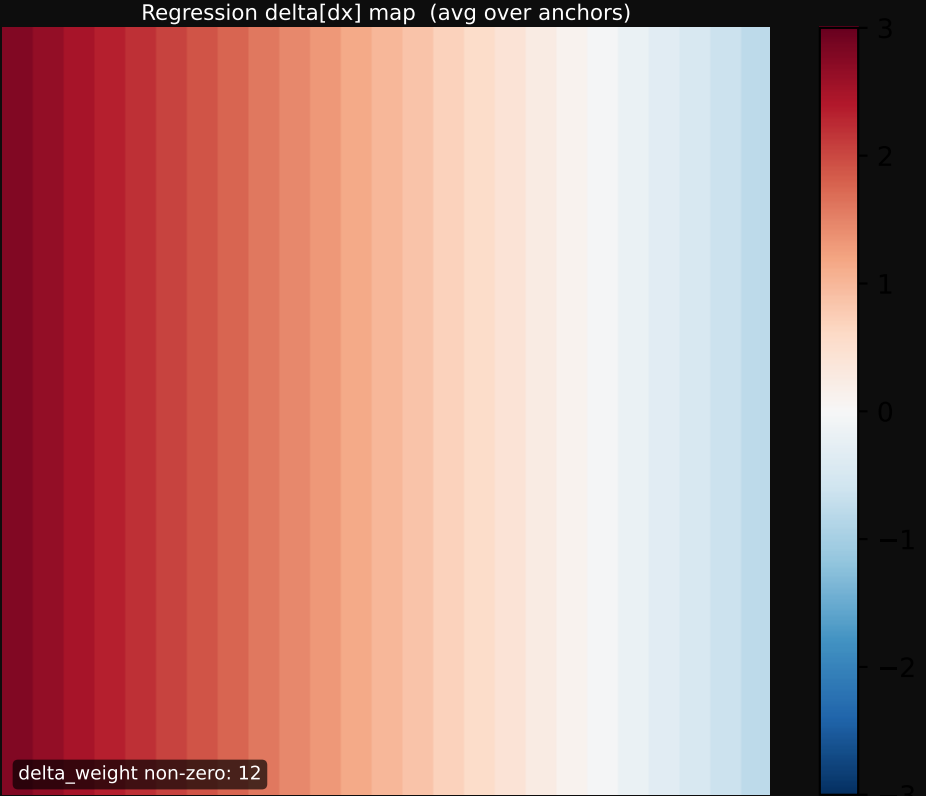
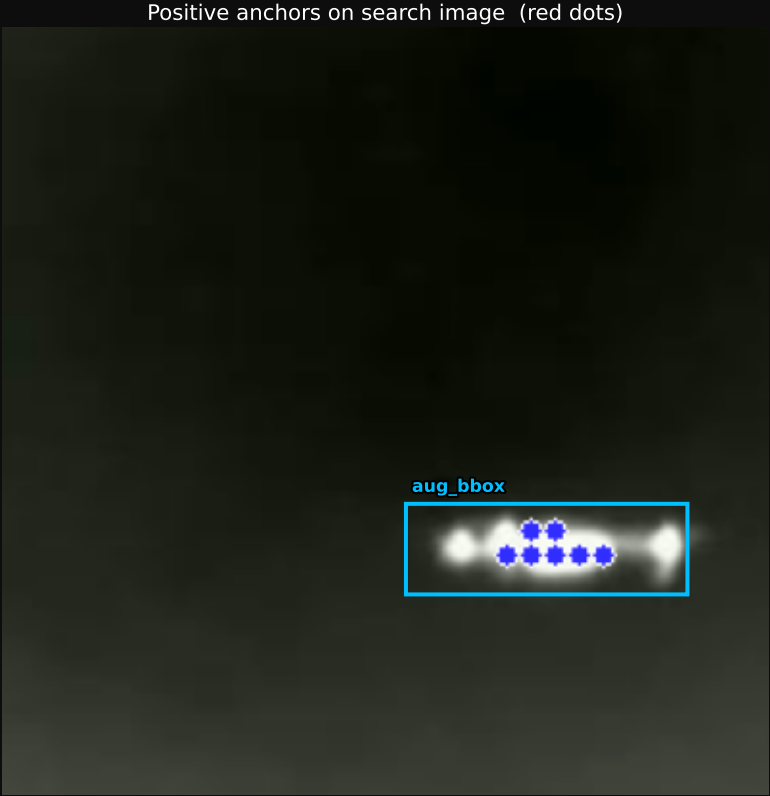
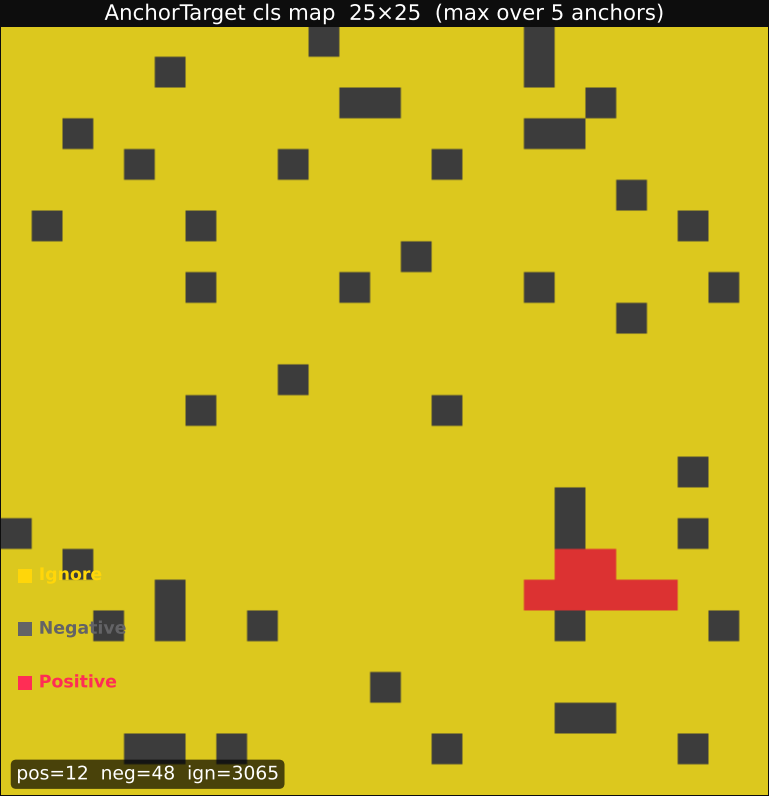


```
— Augmentation config —
TEMPLATE:
SHIFT=4px
SCALE=±5%
BLUR=0.0
FLIP=0.0

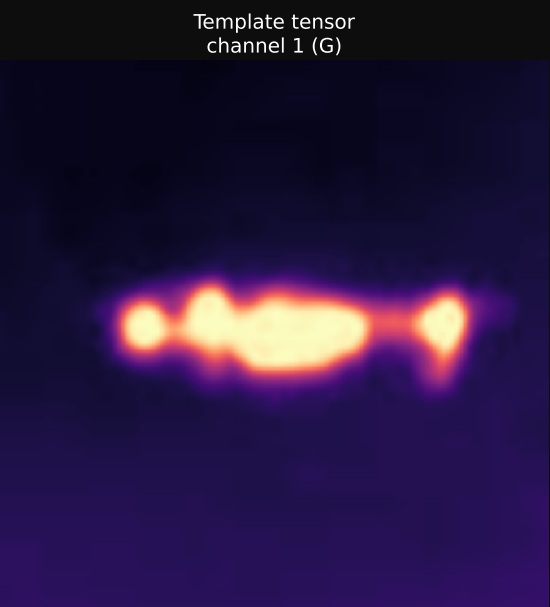
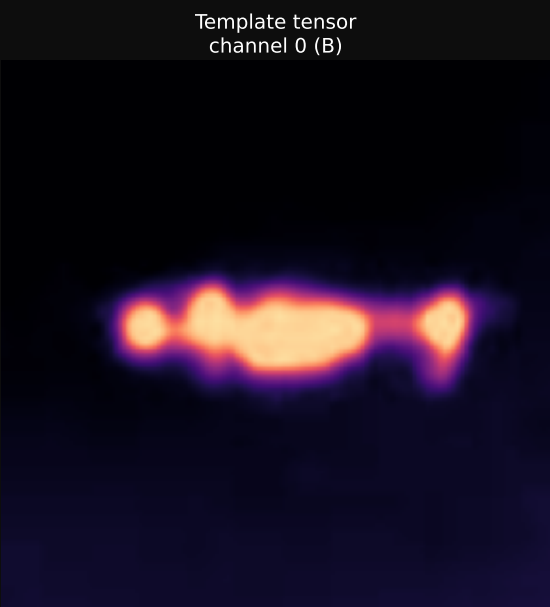
SEARCH:
SHIFT=64px
SCALE=±18%
BLUR=0.0
FLIP=0.0

— aug_bbox (to anchor target) —
x1=133.80 y1=157.76
x2=227.39 y2=187.89
w=93.59
h=30.13
cx=180.59
cy=172.83
```

STEP 3 – ANCHOR TARGET



STEP 4 – MODEL TENSORS



```
— Template tensor (→ model) —
shape : (3, 127, 127)
dtype : float32
min : -10.600
max : 258.375
mean : 35.061
std : 51.425
range : 268.976

— Search tensor (→ model) —
shape : (3, 255, 255)
dtype : float32
min : -2.773
max : 255.834
mean : 28.701
std : 29.861
range : 258.607

— Note —
Raw 0-255 BGR float32
NO /255, NO mean-sub
(matches training-time
PySOT convention)
```

● GT annotation (raw frame)

● Annotation after center-crop / aug_bbox

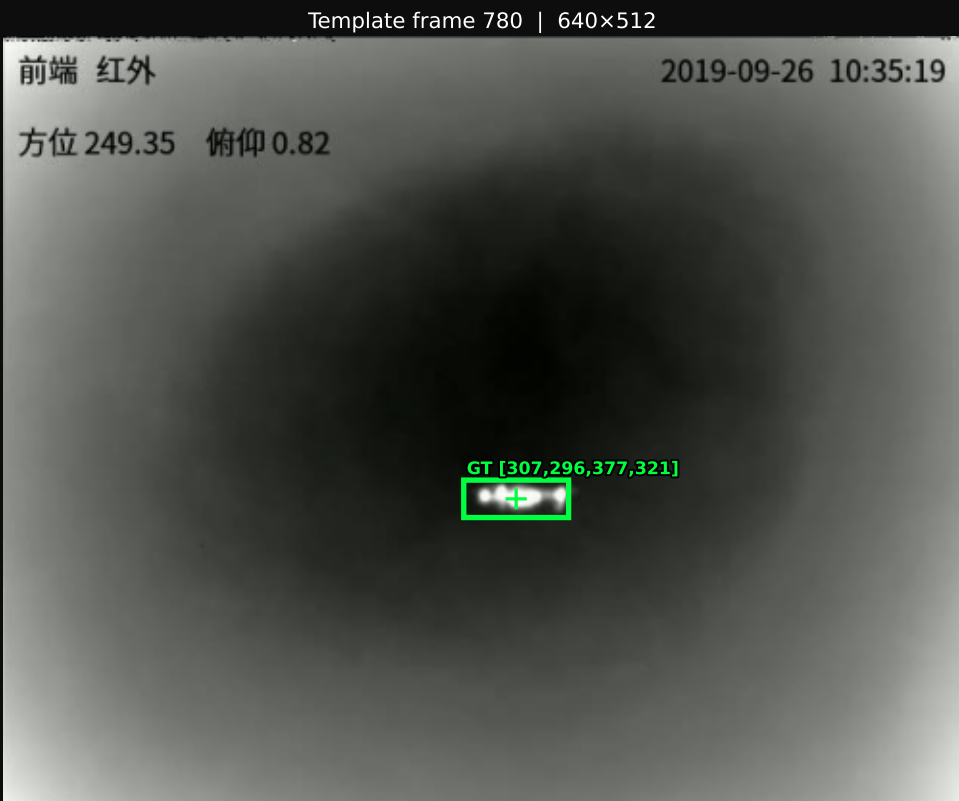
● _get_bbox output (scaled to aug input)

● aug_bbox → anchor_target input

● Image centre crosshair

● Positive anchor

STEP 0 – RAW INPUT



```
— TEMPLATE ANNOTATION —
Format : [x1,y1,x2,y2]
Values : [307.0, 296.0, 377.0, 321.0]
Size : 70.0 x 25.0 px
Center : (342.0, 308.5)
AR : 2.89
Image : 640x512

— SEARCH ANNOTATION —
Values : [315.0, 324.0, 384.0, 349.0]
Size : 69.0 x 25.0 px
Center : (349.5, 336.5)
Image : 640x512
```

STEP 1 – CENTER CROP

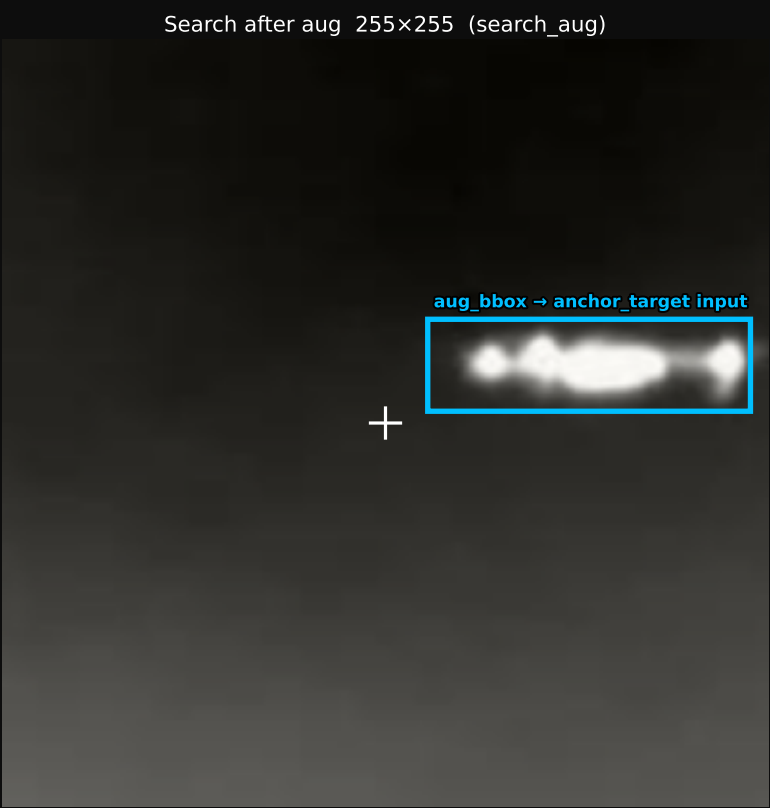
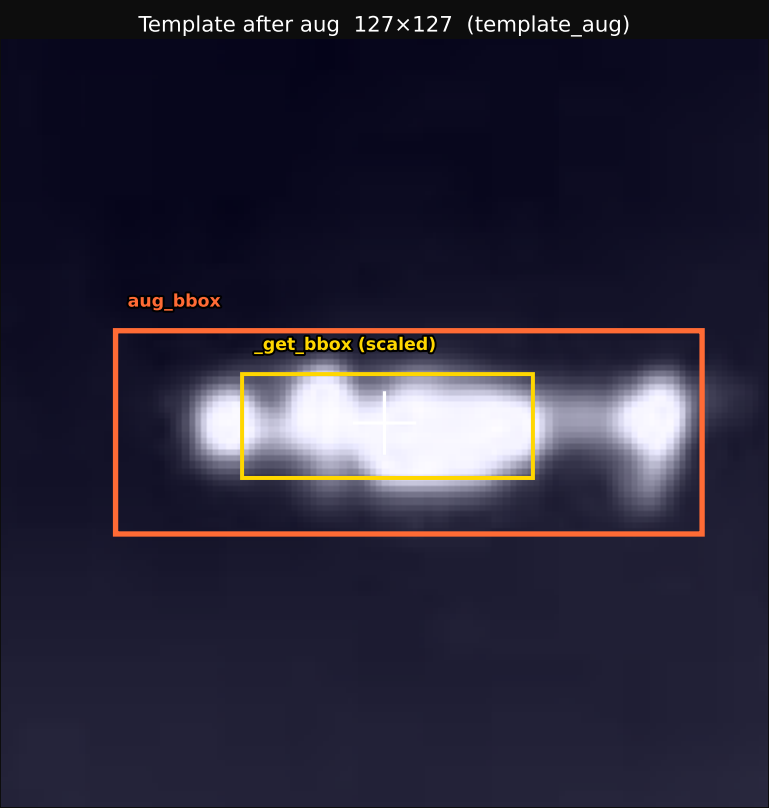


```
— get_center_crop —
Output size (tmpl): 254x254
Output size (srch): 510x510
White cross = image centre
Orange box = new annotation

— get_bbox_output —
TEMPLATE (corner):
[78.8, 109.8, 175.2, 144.2]
cx=127.0 cy=127.0
w=96.3 h=34.4

SEARCH (corner):
[207.1, 237.6, 302.9, 272.4]
cx=255.0 cy=255.0
w=95.9 h=34.7
```

STEP 2 – AUGMENTATION

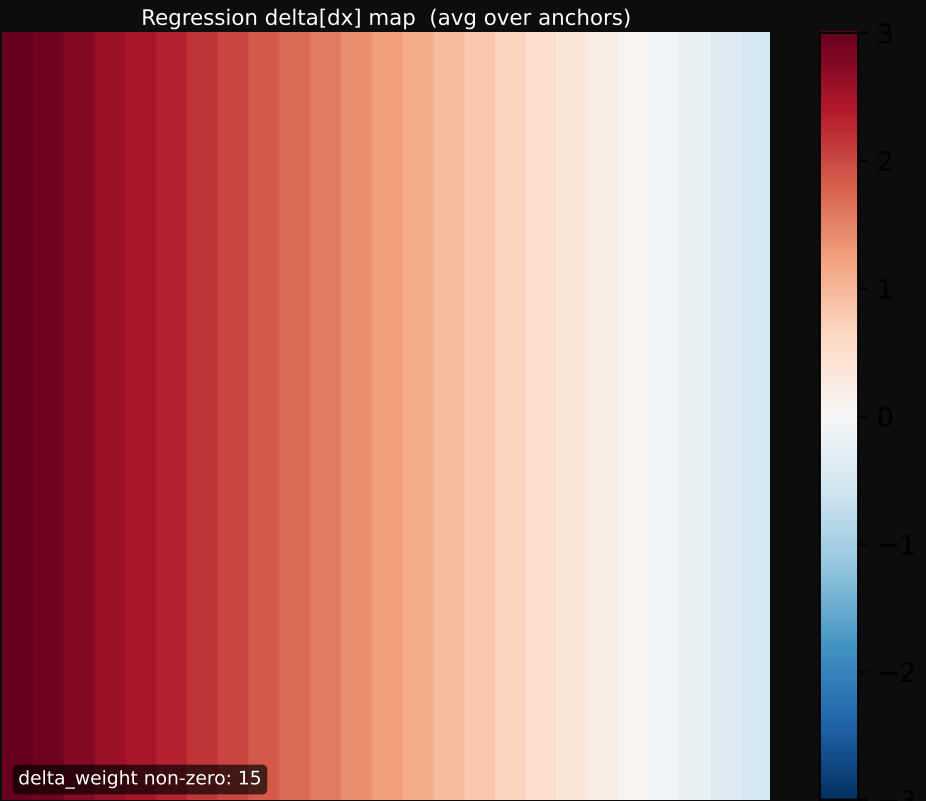
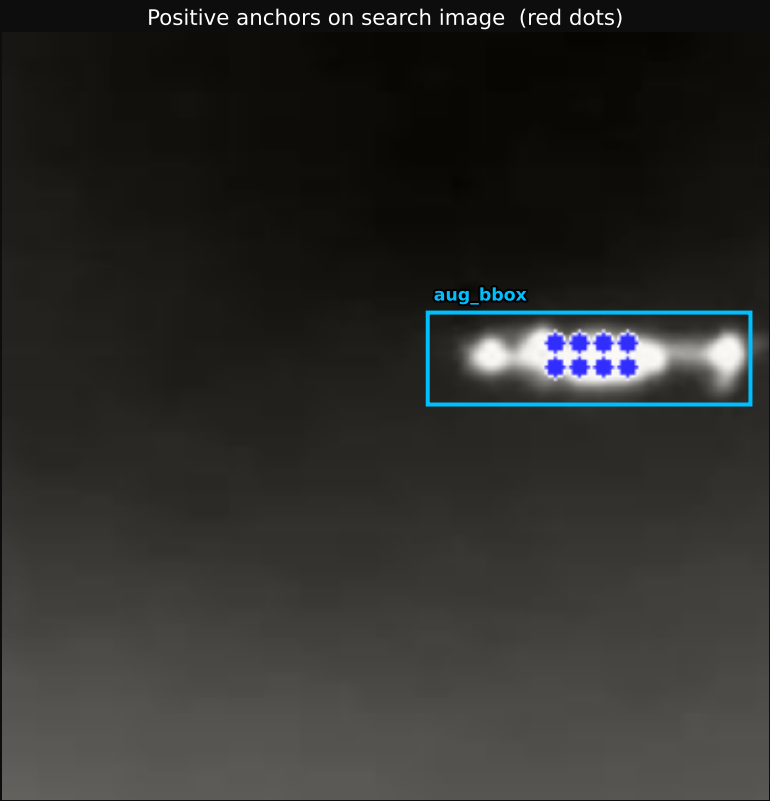
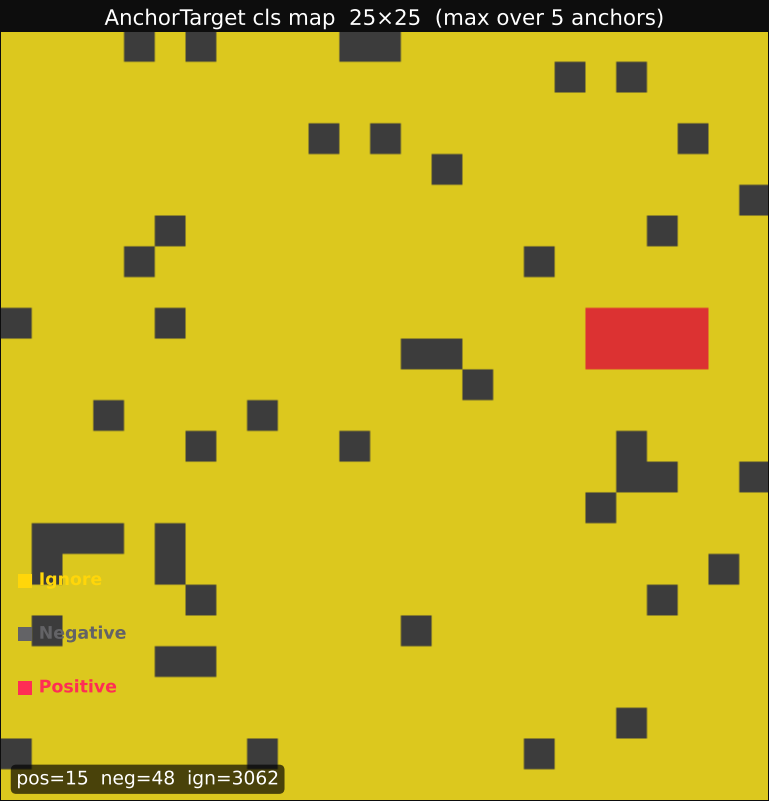


```
— Augmentation config —
TEMPLATE:
SHIFT=4px
SCALE=±5%
BLUR=0.0
FLIP=0.0

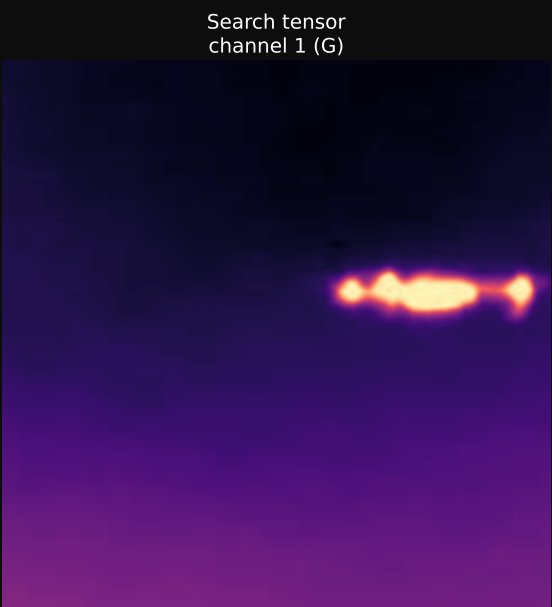
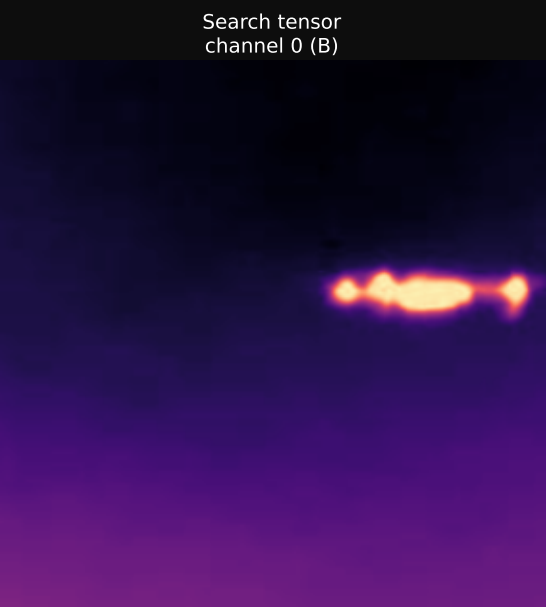
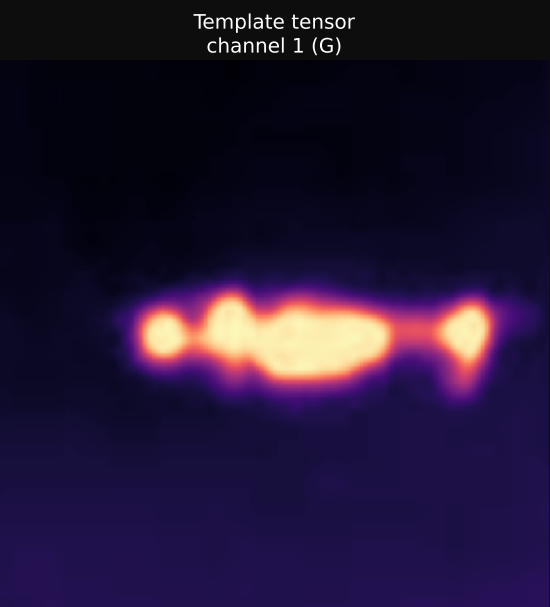
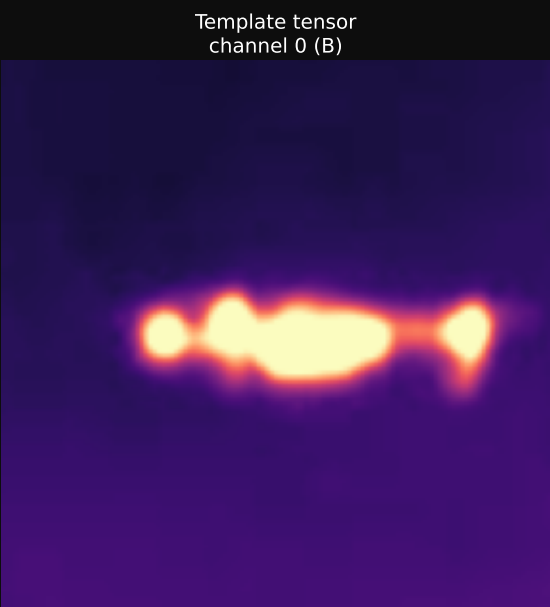
SEARCH:
SHIFT=64px
SCALE=±18%
BLUR=0.0
FLIP=0.0

— aug_bbox (to anchor target) —
x1=141.05 y1=92.50
x2=248.35 y2=123.08
w=107.30
h=30.58
cx=194.70
cy=107.79
```

STEP 3 – ANCHOR TARGET



STEP 4 – MODEL TENSORS



```
— Template tensor (→ model) —
shape : (3, 127, 127)
dtype : float32
min : 3.908
max : 275.316
mean : 45.381
std : 52.241
range : 271.608

— Search tensor (→ model) —
shape : (3, 255, 255)
dtype : float32
min : 1.102
max : 254.808
mean : 45.744
std : 34.608
range : 253.698

— Note —
Raw 0-255 BGR float32
NO /255, NO mean-sub
(matches training-time
PySOT convention)
```

● GT annotation (raw frame)

● Annotation after center-crop / aug_bbox

● _get_bbox output (scaled to aug input)

● aug_bbox → anchor_target input

● Image centre crosshair

● Positive anchor

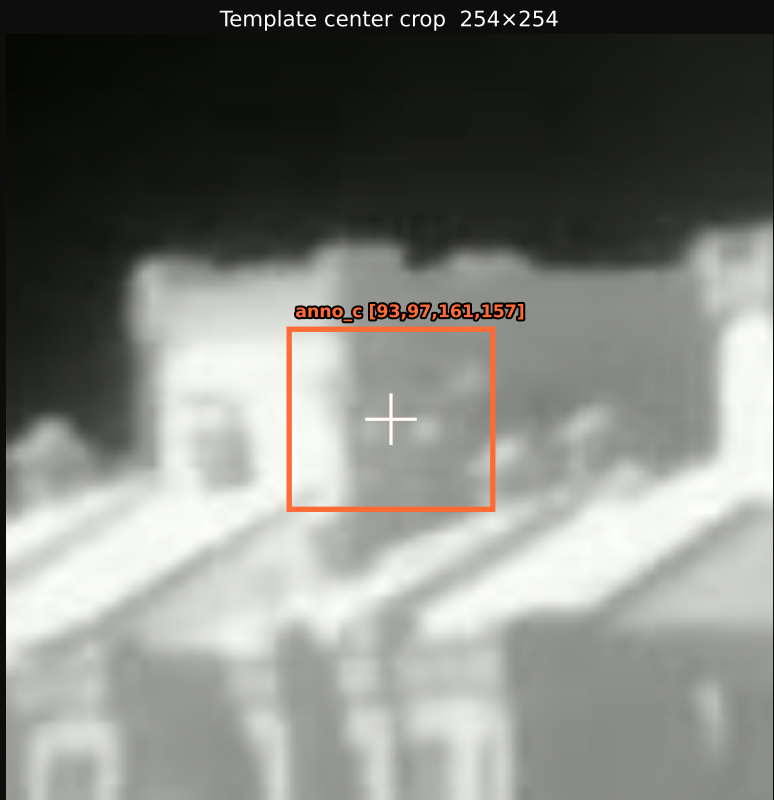
STEP 0 – RAW INPUT



```
— TEMPLATE ANNOTATION —
Format : [x1,y1,x2,y2]
Values : [442.0, 262.0, 468.0, 285.0]
Size : 26.0 x 23.0 px
Center : (455.0, 273.5)
AR : 1.13
Image : 640x512

— SEARCH ANNOTATION —
Values : [460.0, 265.0, 490.0, 287.0]
Size : 30.0 x 22.0 px
Center : (479.5, 276.0)
Image : 640x512
```

STEP 1 – CENTER CROP

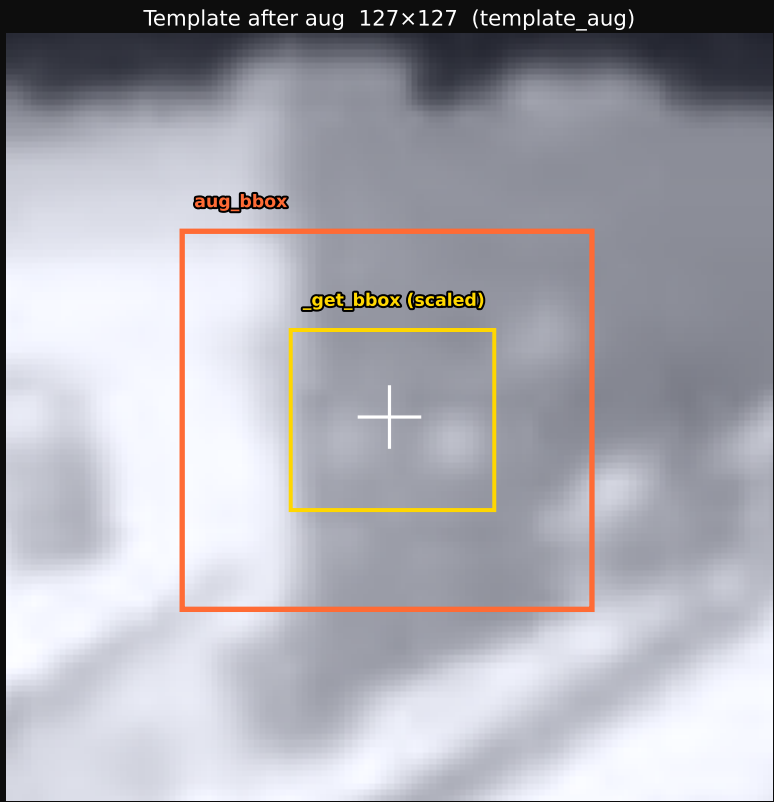


```
— get_center_crop —
Output size (tmpl): 254x254
Output size (srch): 510x510
White cross = image centre
Orange box = new annotation

— get_bbox_output —
TEMPLATE (corner):
[93.3, 97.2, 168.7, 156.8]
cx=127.0 cy=127.0
w=67.4 h=59.6

SEARCH (corner):
[214.0, 231.9, 296.0, 278.1]
cx=255.0 cy=255.0
w=82.0 h=46.3
```

STEP 2 – AUGMENTATION

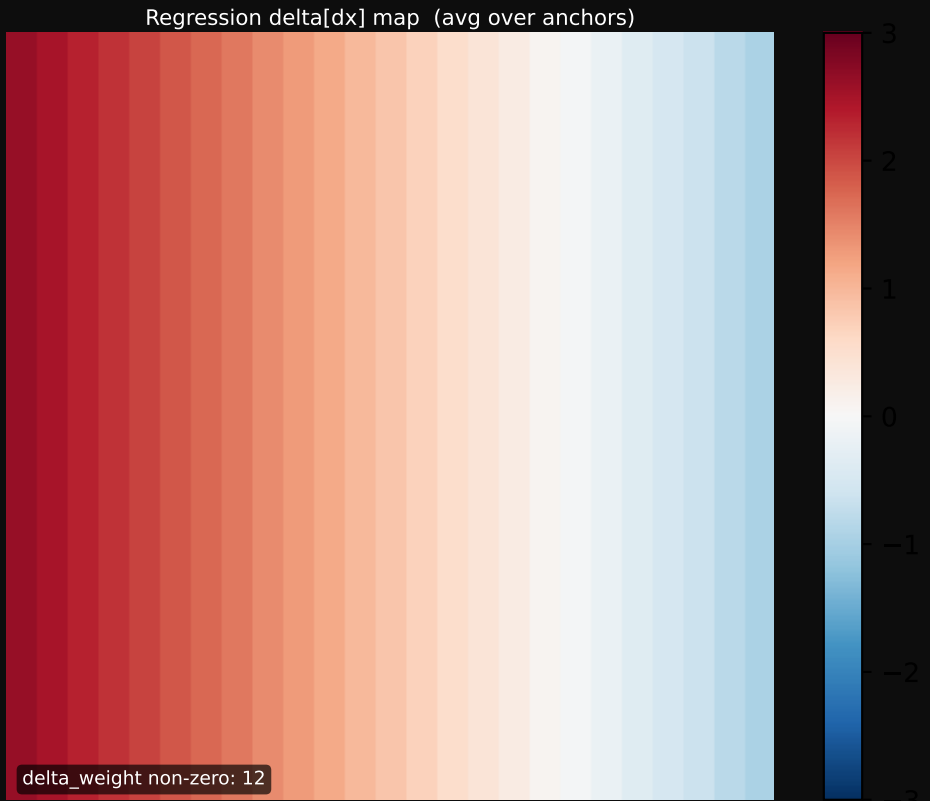


```
— Augmentation config —
TEMPLATE:
SHIFT=4px
SCALE=5%
BLUR=0.0
FLIP=0.0

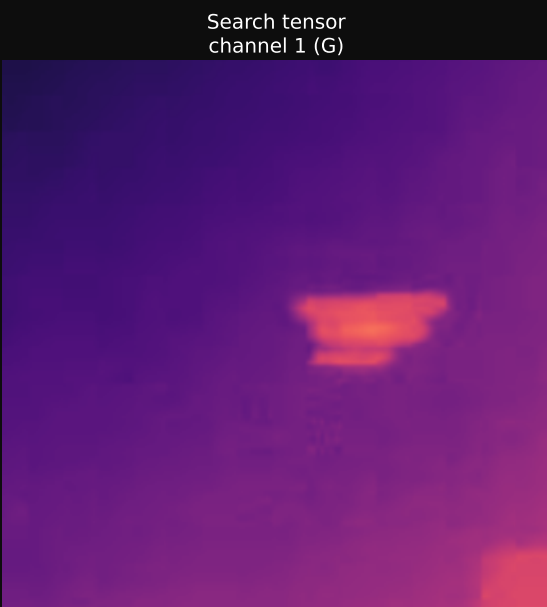
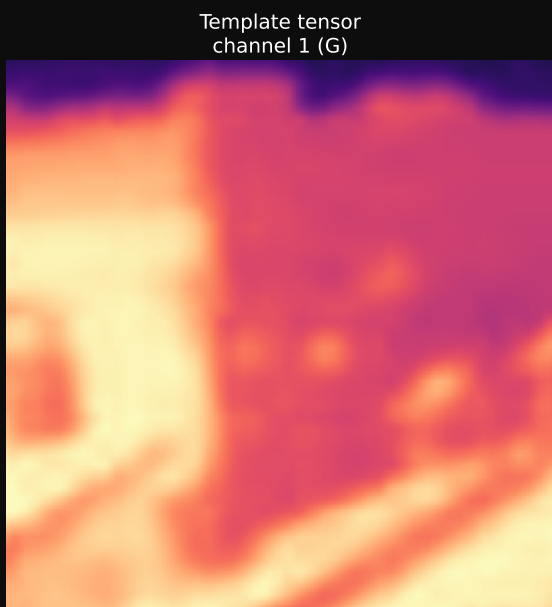
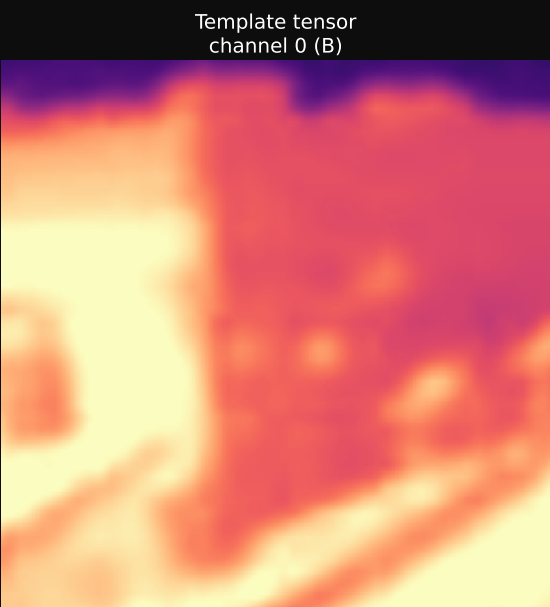
SEARCH:
SHIFT=64px
SCALE=18%
BLUR=0.0
FLIP=0.0

— aug_bbox (to anchor target) —
x1=132.53 y1=102.39
x2=211.10 y2=148.17
w=78.57
h=45.77
cx=171.82
cy=125.28
```

STEP 3 – ANCHOR TARGET



STEP 4 – MODEL TENSORS



```
— Template tensor (→ model) —
shape : (3, 127, 127)
dtype : float32
min : 34.375
max : 265.244
mean : 182.612
std : 49.986
range : 230.869

— Search tensor (→ model) —
shape : (3, 255, 255)
dtype : float32
min : 28.686
max : 179.075
mean : 74.383
std : 24.311
range : 158.389

— Note —
Raw 0-255 BGR float32
NO /255, NO mean-sub
(matches training-time
PySOT convention)
```

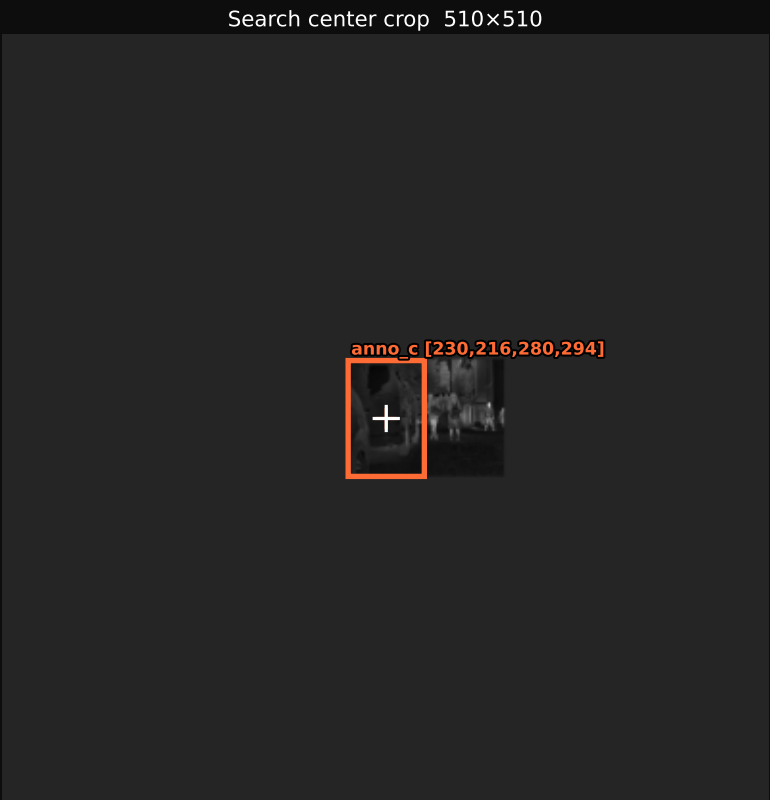
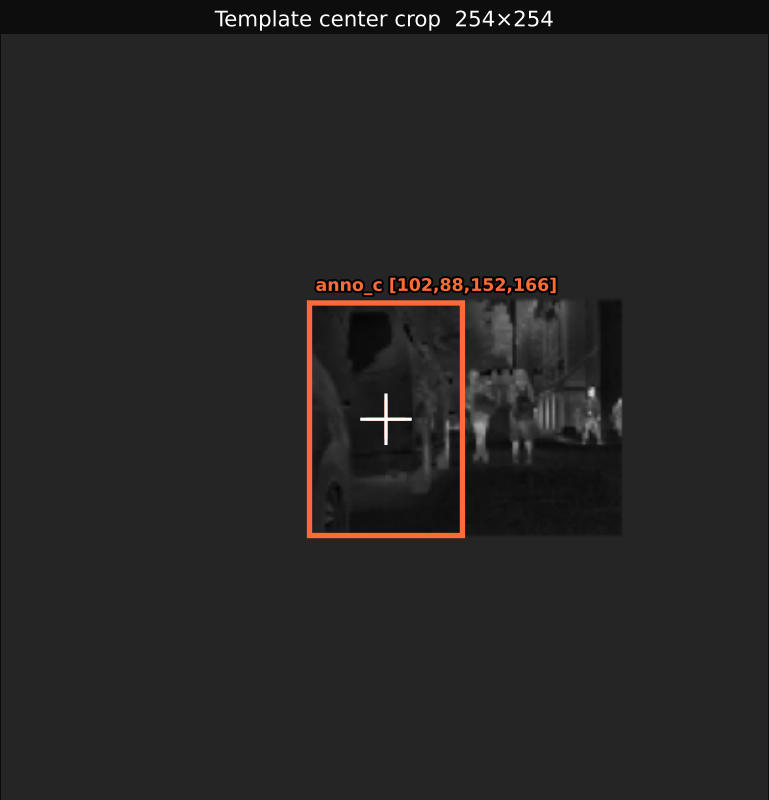
STEP 0 – RAW INPUT



```
— TEMPLATE ANNOTATION —
Format : [x1,y1,x2,y2]
Values : [0.0, 0.0, 315.0, 479.0]
Size : 315.0 × 479.0 px
Center : (157.5, 239.5)
AR : 0.66
Image : 640×480

— SEARCH ANNOTATION —
Values : [0.0, 0.0, 315.0, 479.0]
Size : 315.0 × 479.0 px
Center : (157.5, 239.5)
Image : 640×480
```

STEP 1 – CENTER CROP

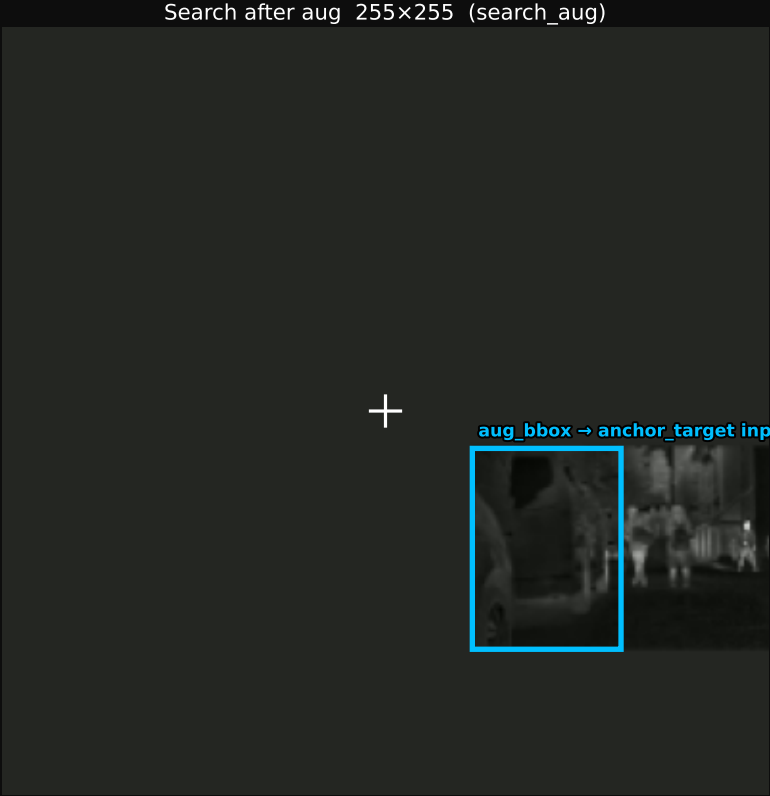


```
— get_center_crop —
Output size (tmpl): 254×254
Output size (srch): 510×510
White cross = image centre
Orange box = new annotation

— get_bbox_output —
TEMPLATE (corner):
[101.7, 88.5, 152.3, 165.5]
cx=127.0 cy=127.0
w=50.7 h=77.0

SEARCH (corner):
[229.7, 216.5, 280.3, 293.5]
cx=255.0 cy=255.0
w=50.7 h=77.0
```

STEP 2 – AUGMENTATION

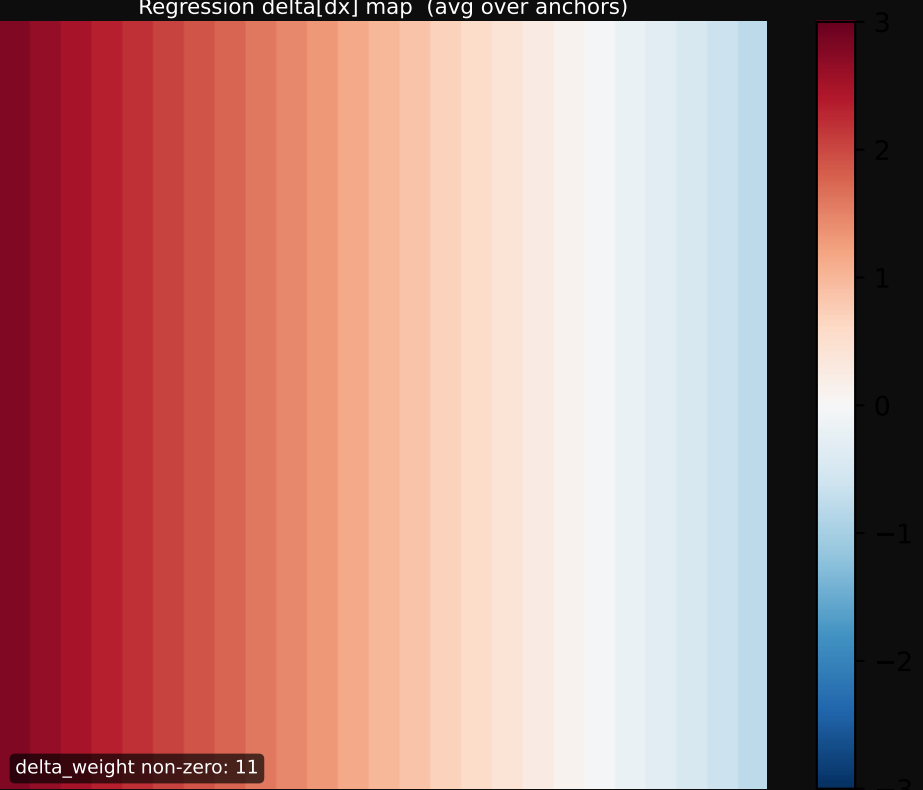
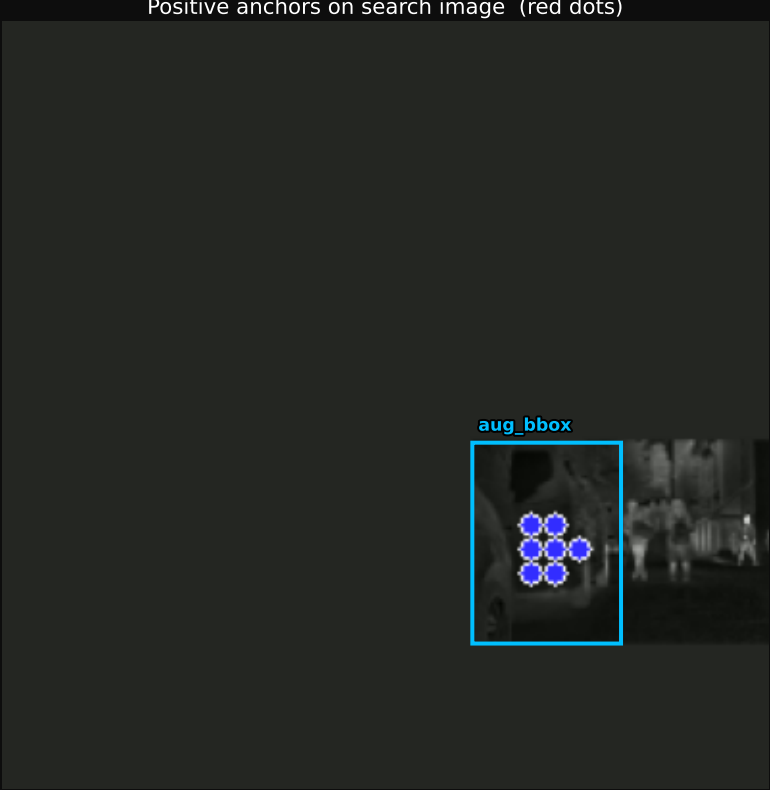
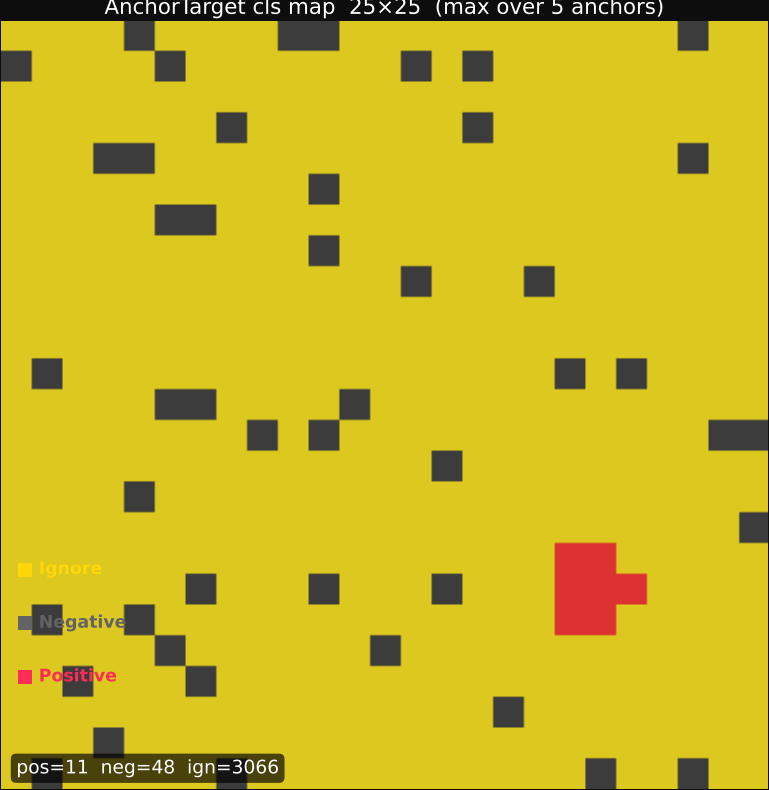


```
— Augmentation config —
TEMPLATE:
SHIFT=4px
SCALE=±5%
BLUR=0.0
FLIP=0.0

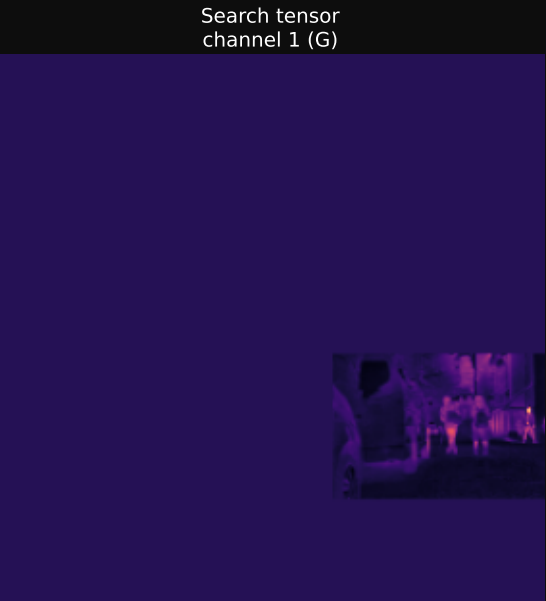
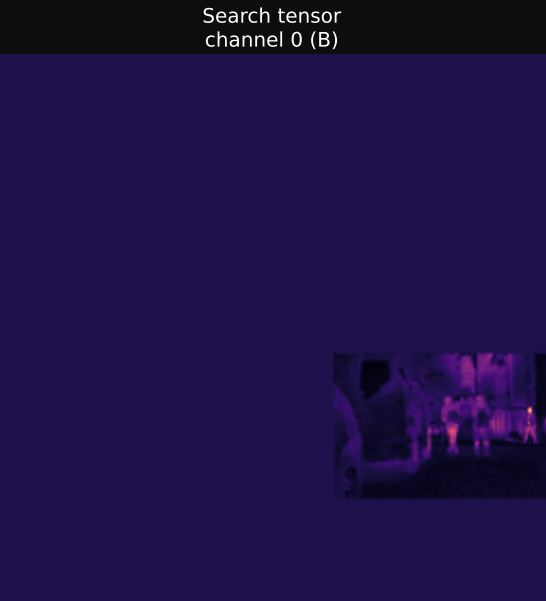
SEARCH:
SHIFT=64px
SCALE=±10%
BLUR=0.0
FLIP=0.0

— aug_bbox (to anchor target) —
x1=155.87 y1=139.43
x2=205.32 y2=206.22
w=49.44
h=66.79
cx=180.59
cy=172.83
```

STEP 3 – ANCHOR TARGET



STEP 4 – MODEL TENSORS



```
— Template tensor (→ model) —
shape : (3, 127, 127)
dtype : float32
min : -8.600
max : 170.375
mean : 33.709
std : 15.107
range : 178.976

— Search tensor (→ model) —
shape : (3, 255, 255)
dtype : float32
min : 1.227
max : 222.834
mean : 36.470
std : 7.016
range : 221.607

— Note —
Raw 0-255 BGR float32
NO /255, NO mean-sub
(matches training-time
PySOT convention)
```

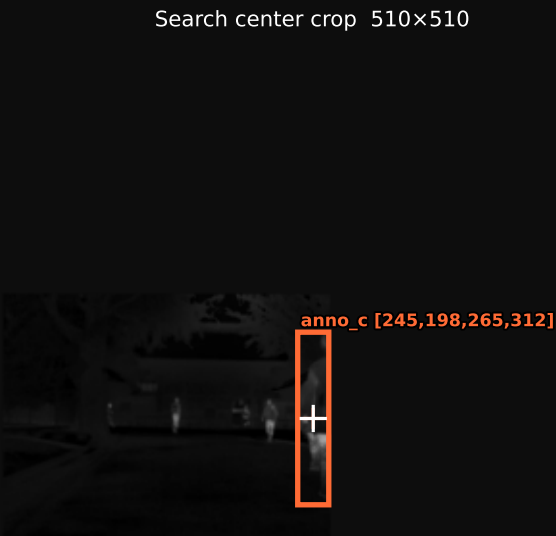
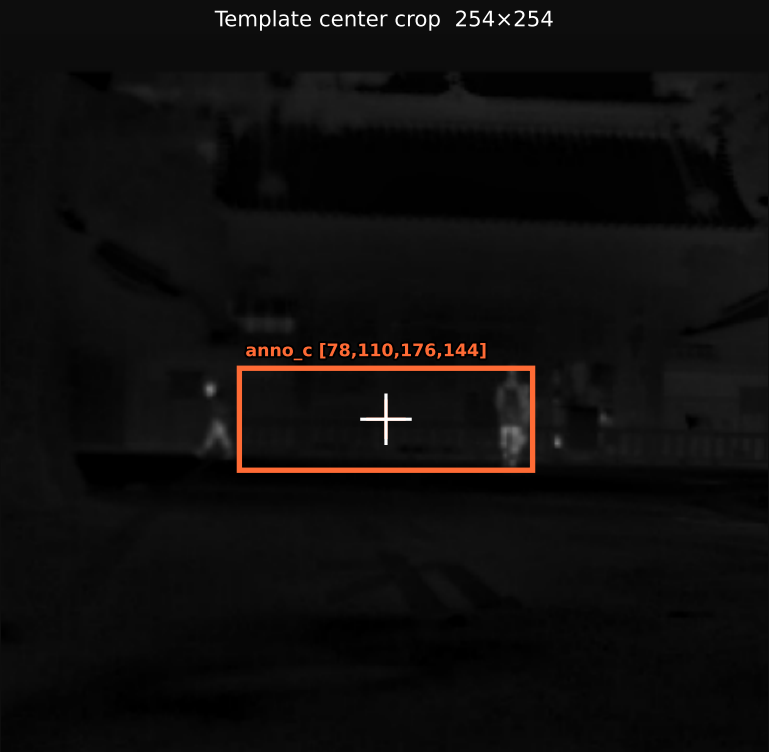
STEP 0 – RAW INPUT



```
— TEMPLATE ANNOTATION —
Format : [x1,y1,x2,y2]
Values : [239.0, 208.0, 446.0, 280.0]
Size : 207.0 × 72.0 px
Center : (342.5, 244.0)
AR : 2.88
Image : 640×480

— SEARCH ANNOTATION —
Values : [577.0, 74.0, 638.0, 413.0]
Size : 61.0 × 339.0 px
Center : (607.5, 243.5)
Image : 640×480
```

STEP 1 – CENTER CROP

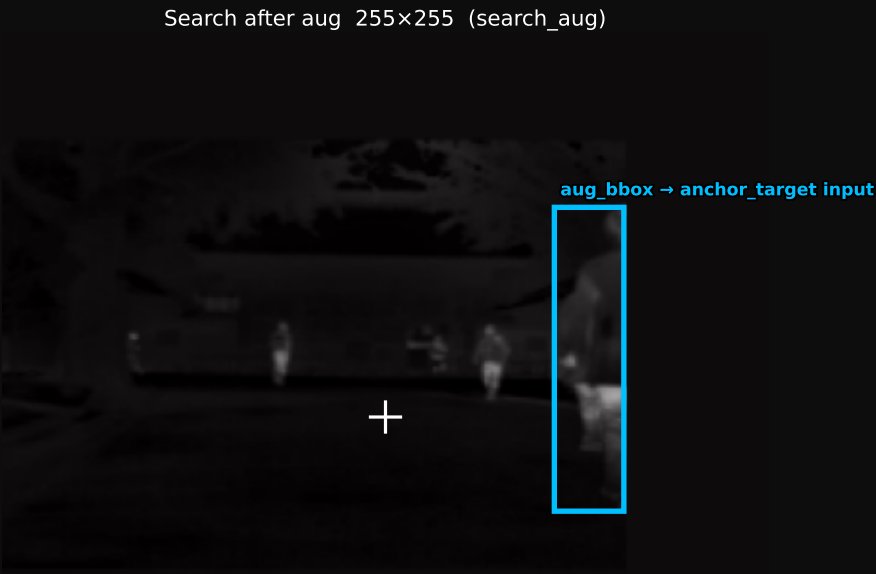
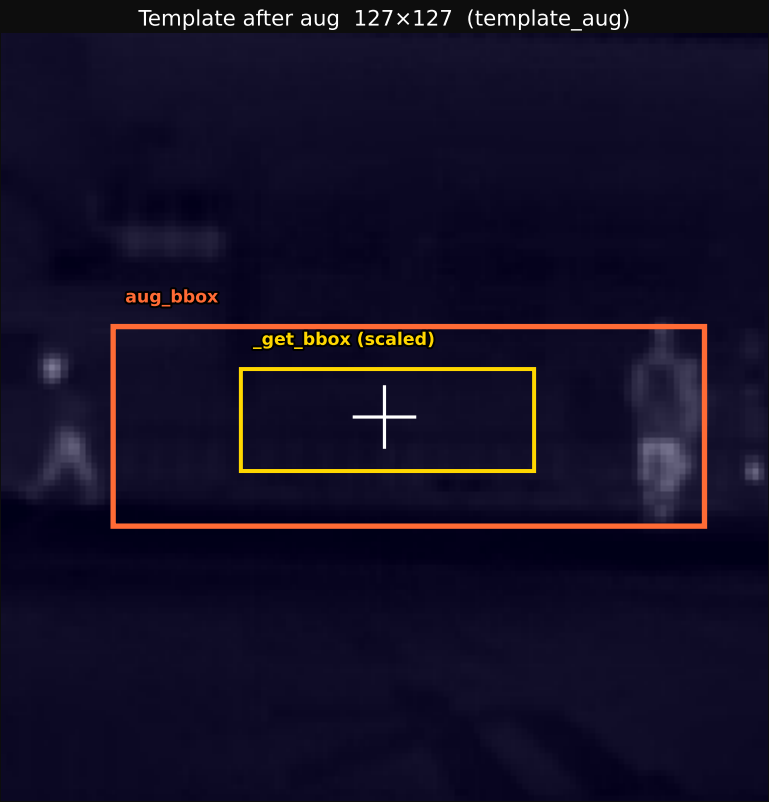


```
— get_center_crop —
Output size (tmpl): 254×254
Output size (srch): 510×510
White cross = image centre
Orange box = new annotation

— get_bbox_output —
TEMPLATE (corner):
[78.4, 110.1, 175.6, 143.9]
cx=127.0 cy=127.0
w=97.1 h=33.8

SEARCH (corner):
[244.7, 197.6, 265.3, 312.4]
cx=255.0 cy=255.0
w=20.7 h=114.8
```

STEP 2 – AUGMENTATION

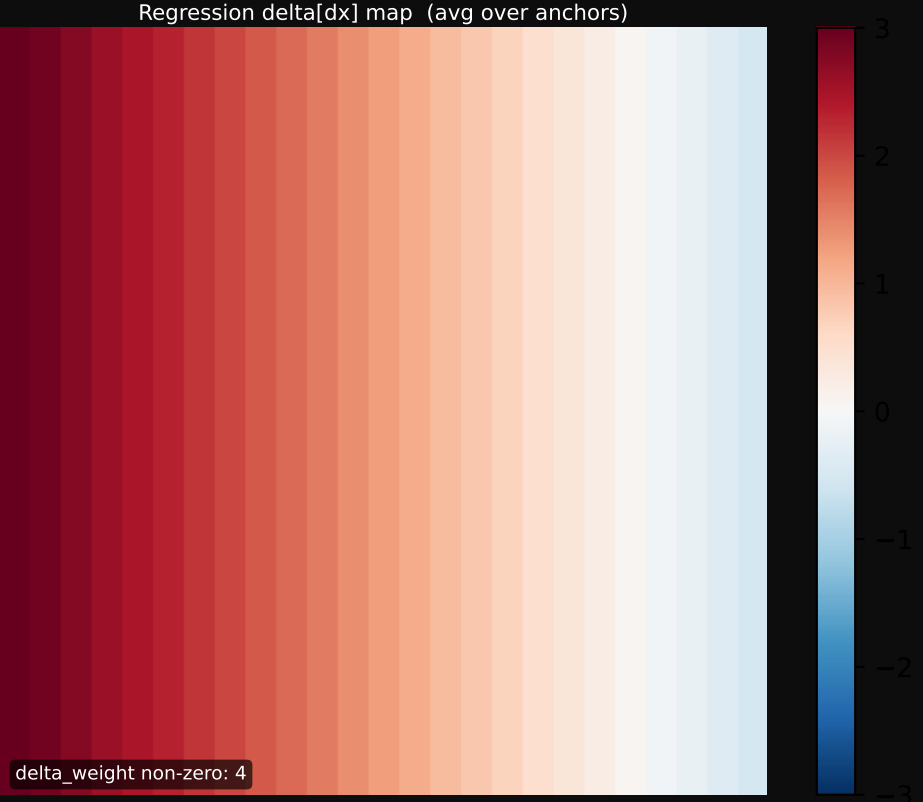
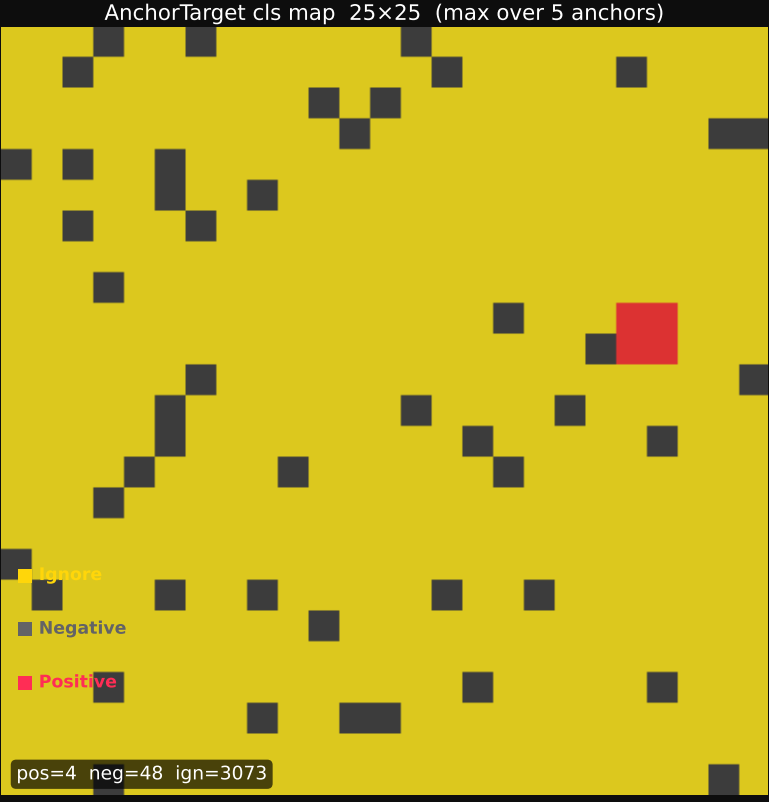


```
— Augmentation config —
TEMPLATE:
SHIFT=4px
SCALE=±5%
BLUR=0.0
FLIP=0.0

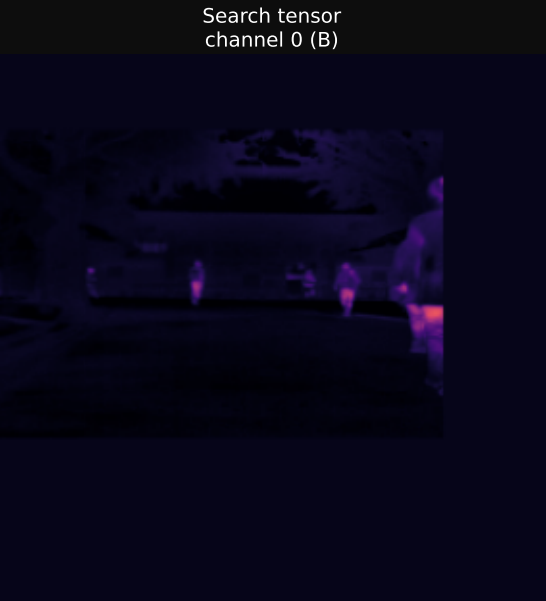
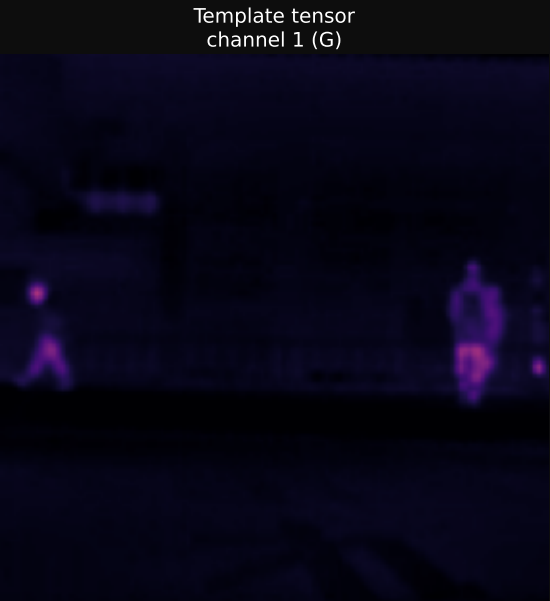
SEARCH:
SHIFT=64px
SCALE=±18%
BLUR=0.0
FLIP=0.0

— aug_bbox (to anchor target) —
x1=183.14 y1=57.28
x2=206.25 y2=158.31
w=23.11
h=101.03
cx=194.70
cy=107.79
```

STEP 3 – ANCHOR TARGET



STEP 4 – MODEL TENSORS



```
— Template tensor (→ model) —
shape : (3, 127, 127)
dtype : float32
min : -2.092
max : 149.516
mean : 21.480
std : 14.816
range : 151.608

— Search tensor (→ model) —
shape : (3, 255, 255)
dtype : float32
min : -1.402
max : 168.800
mean : 13.261
std : 8.154
range : 170.201

— Note —
Raw 0-255 BGR float32
NO /255, NO mean-sub
(matches training-time
PySOT convention)
```

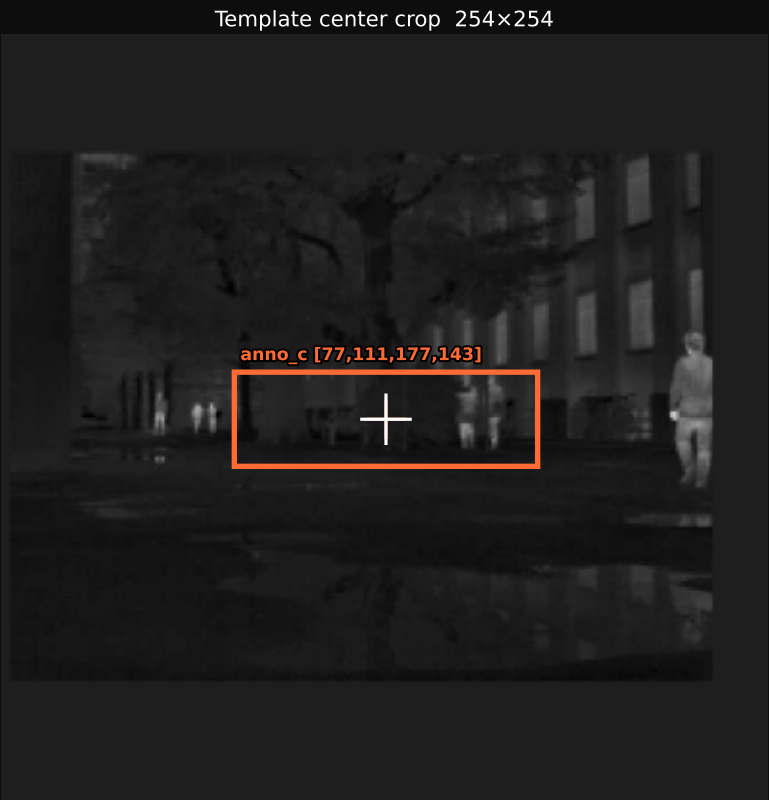
STEP 0 - RAW INPUT



```
— TEMPLATE ANNOTATION —
Format : [x1,y1,x2,y2]
Values : [204.0, 198.0, 481.0, 284.0]
Size : 277.0 x 86.0 px
Center : (342.5, 241.0)
AR : 3.22
Image : 640x480

— SEARCH ANNOTATION —
Values : [390.0, 207.0, 515.0, 261.0]
Size : 125.0 x 54.0 px
Center : (452.5, 234.0)
Image : 640x480
```

STEP 1 - CENTER CROP

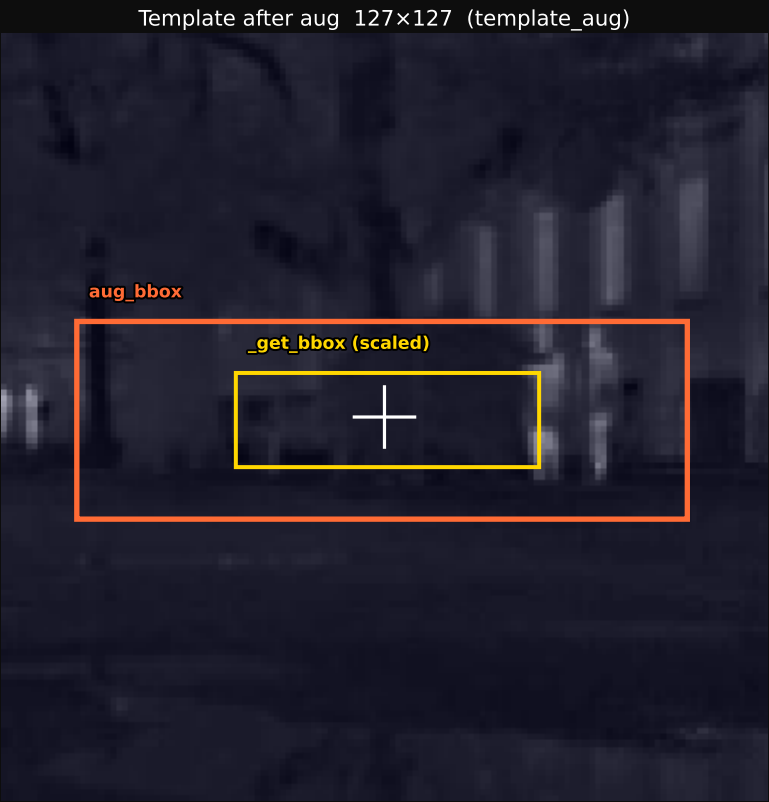


```
— get_center_crop —
Output size (tmpl): 254x254
Output size (srch): 510x510
White cross = image centre
Orange box = new annotation

— get_bbox_output —
TEMPLATE (corner):
[76.8, 111.4, 177.2, 142.6]
cx=127.0 cy=127.0
w=100.5 h=31.2

SEARCH (corner):
[209.8, 235.5, 300.2, 274.5]
cx=255.0 cy=255.0
w=90.5 h=39.1
```

STEP 2 - AUGMENTATION

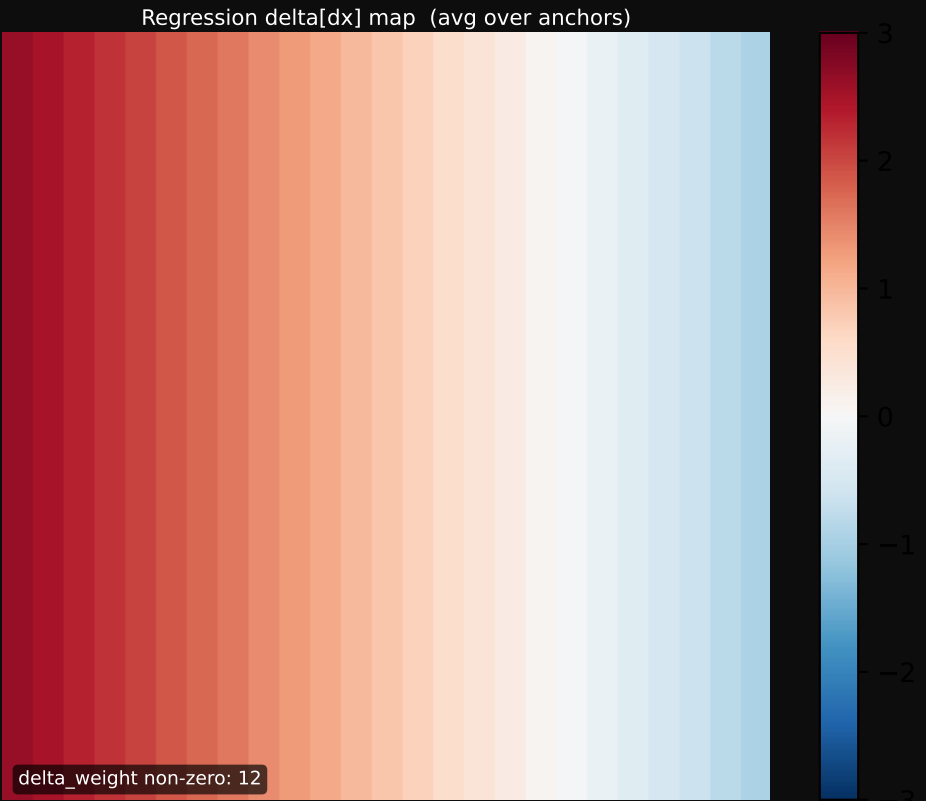
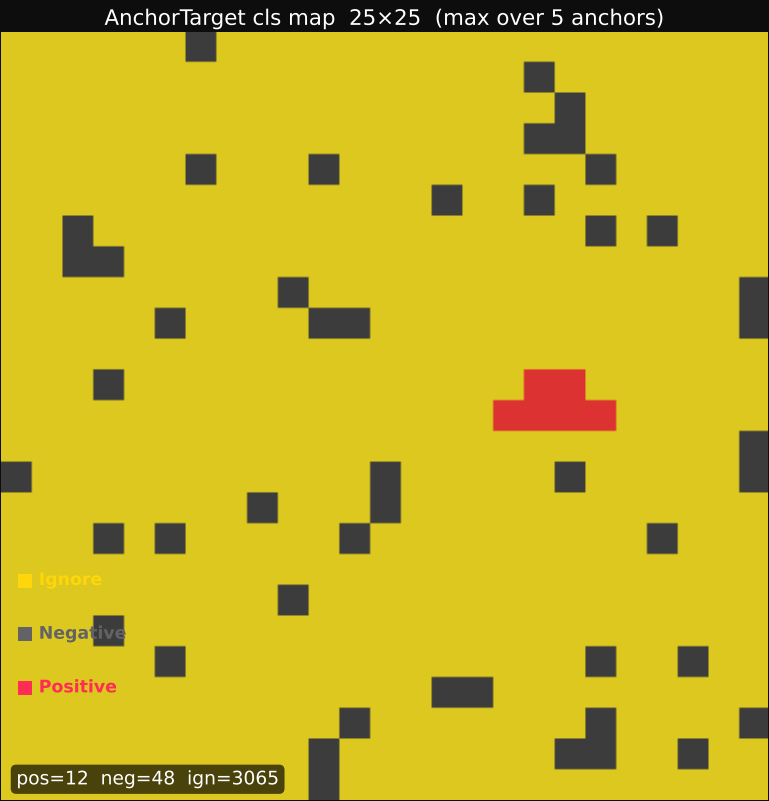


```
— Augmentation config —
TEMPLATE:
SHIFT=4px
SCALE=5%
BLUR=0.0
FLIP=0.0

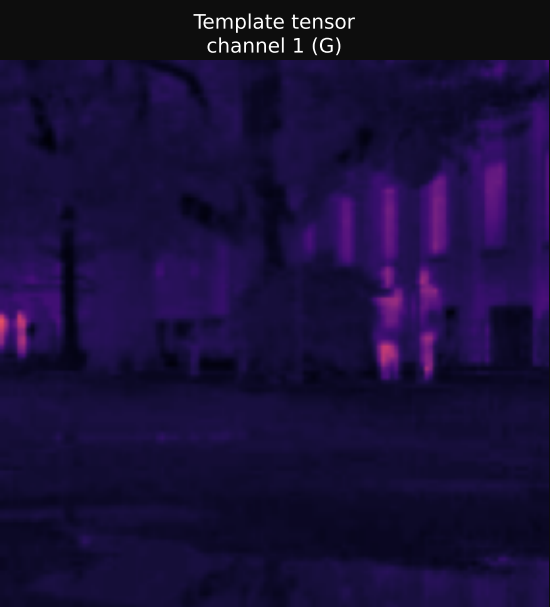
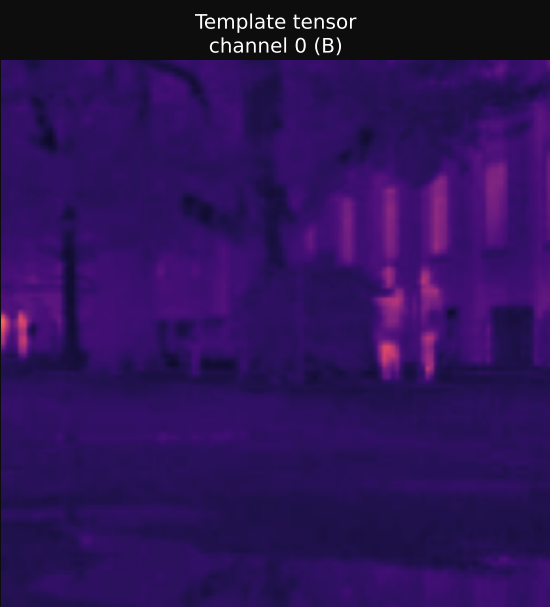
SEARCH:
SHIFT=64px
SCALE=10%
BLUR=0.0
FLIP=0.0

— aug_bbox (to anchor target) —
x1=128.47 y1=105.94
x2=215.17 y2=144.62
w=86.70
h=38.68
cx=171.82
cy=125.28
```

STEP 3 - ANCHOR TARGET



STEP 4 - MODEL TENSORS



```
— Template tensor (→ model) —
shape : (3, 127, 127)
dtype : float32
min : 3.225
max : 177.244
mean : 34.394
std : 13.936
range : 174.019

— Search tensor (→ model) —
shape : (3, 255, 255)
dtype : float32
min : 9.686
max : 222.191
mean : 57.362
std : 26.759
range : 212.506

— Note —
Raw 0-255 BGR float32
NO /255, NO mean-sub
(matches training-time
PySOT convention)
```

STEP 0 - RAW INPUT



```
— TEMPLATE ANNOTATION —
Format : [x1,y1,x2,y2]
Values : [334.0, 189.0, 409.0, 223.0]
Size : 75.0 x 34.0 px
Center : (371.5, 206.0)
AR : 2.21
Image : 640x512

— SEARCH ANNOTATION —
Values : [334.0, 189.0, 409.0, 223.0]
Size : 75.0 x 34.0 px
Center : (371.5, 206.0)
AR : 2.21
Image : 640x512
```

STEP 1 - CENTER CROP



```
— get_center_crop —
Output size (tmpl): 254x254
Output size (srch): 510x510
White cross = image centre
Orange box = new annotation

— get_bbox_output —
TEMPLATE (corner):
[82.5, 106.8, 171.5, 147.2]
cx=127.0 cy=127.0
w=89.0 h=40.3

SEARCH (corner):
[210.5, 234.8, 299.5, 275.2]
cx=255.0 cy=255.0
w=89.0 h=40.3
```

STEP 2 - AUGMENTATION

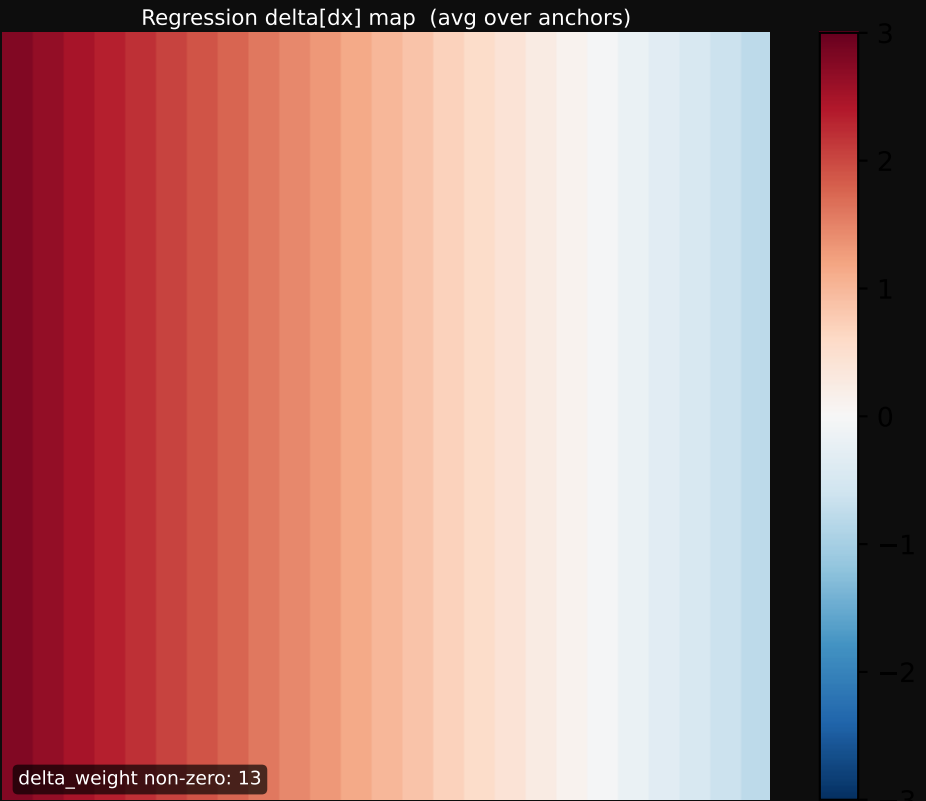
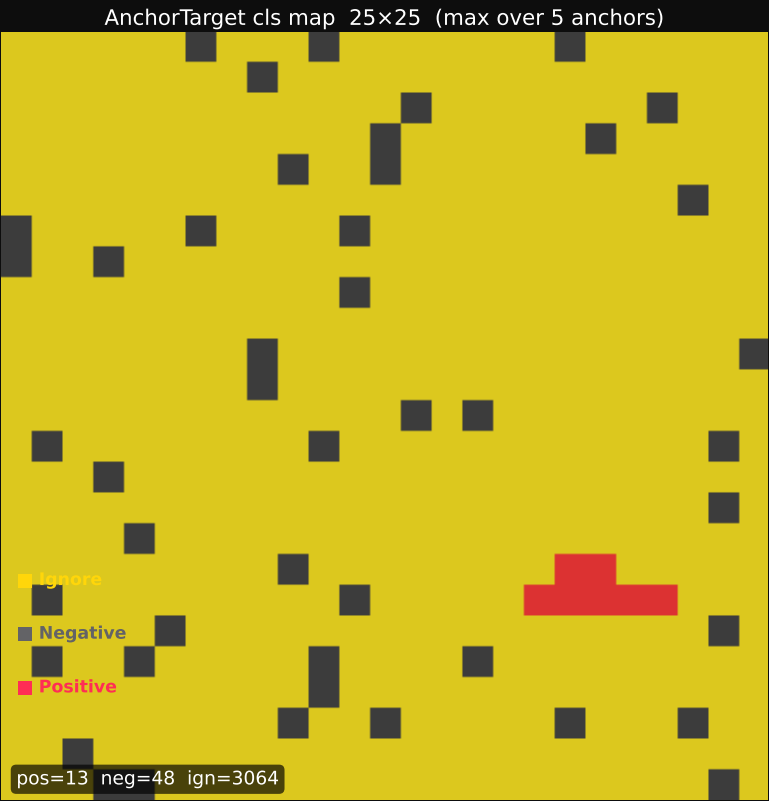


```
— Augmentation config —
TEMPLATE:
SHIFT=4px
SCALE=5%
BLUR=0.0
FLIP=0.0

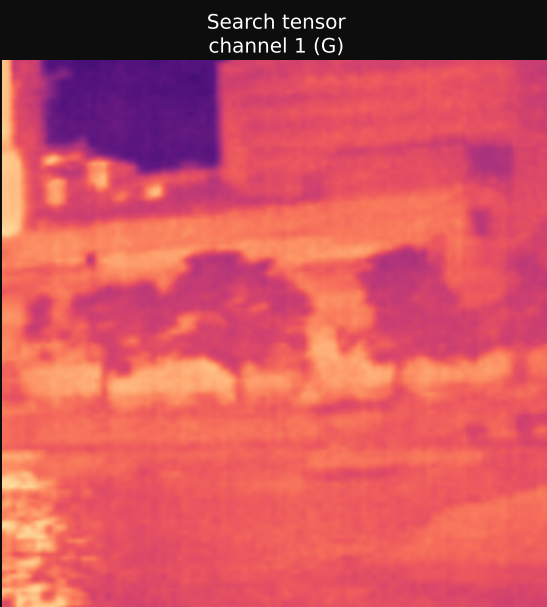
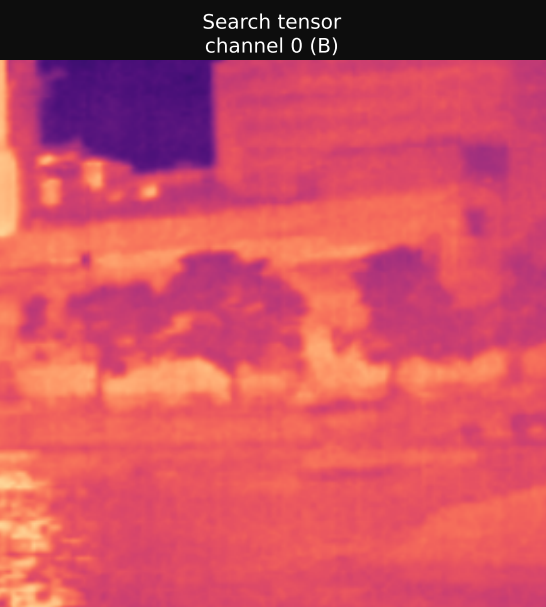
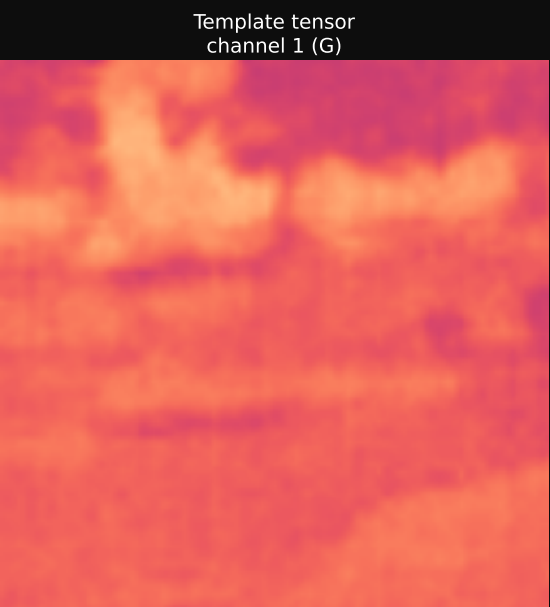
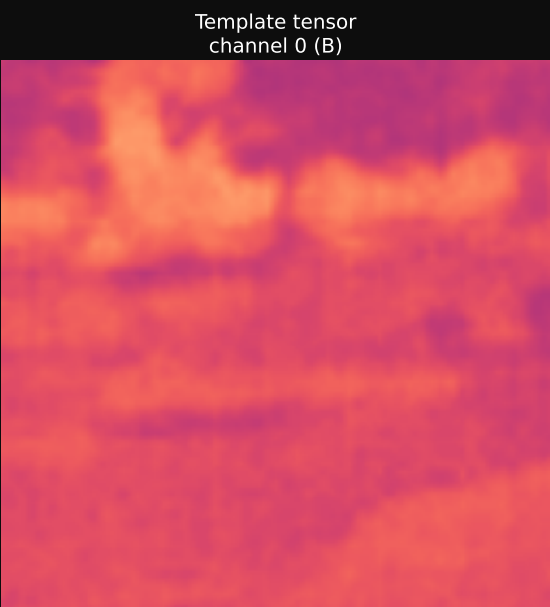
SEARCH:
SHIFT=64px
SCALE=18%
BLUR=0.0
FLIP=0.0

— aug_bbox (to anchor target) —
x1=137.17 y1=155.34
x2=224.02 y2=190.31
w=86.85
h=34.98
cx=180.59
cy=172.83
```

STEP 3 - ANCHOR TARGET



STEP 4 - MODEL TENSORS



```
— Template tensor (-> model) —
shape : (3, 127, 127)
dtype : float32
min : 124.408
max : 216.375
mean : 165.894
std : 15.335
range : 91.976

— Search tensor (-> model) —
shape : (3, 255, 255)
dtype : float32
min : 51.227
max : 237.834
mean : 159.334
std : 28.624
range : 186.607

— Note —
Raw 0-255 BGR float32
NO /255, NO mean-sub
(matches training-time
PySOT convention)
```

STEP 0 – RAW INPUT



```
— TEMPLATE ANNOTATION —
Format : [x1,y1,x2,y2]
Values : [110.0, 156.0, 147.0, 169.0]
Size : 37.0 × 13.0 px
Center : (128.5, 162.5)
AR : 2.85
Image : 640×512

— SEARCH ANNOTATION —
Values : [110.0, 156.0, 147.0, 169.0]
Size : 37.0 × 13.0 px
Center : (128.5, 162.5)
Image : 640×512
```

STEP 1 – CENTER CROP



```
— get_center_crop —
Output size (tmpl): 254×254
Output size (srch): 510×510
White cross = image centre
Orange box = new annotation

— get_bbox_output —
TEMPLATE (corner):
[78.6, 110.0, 175.4, 144.0]
cx=127.0 cy=127.0
w=96.8 h=34.0

SEARCH (corner):
[206.0, 238.0, 303.4, 272.0]
cx=255.0 cy=255.0
w=96.8 h=34.0
```

STEP 2 – AUGMENTATION

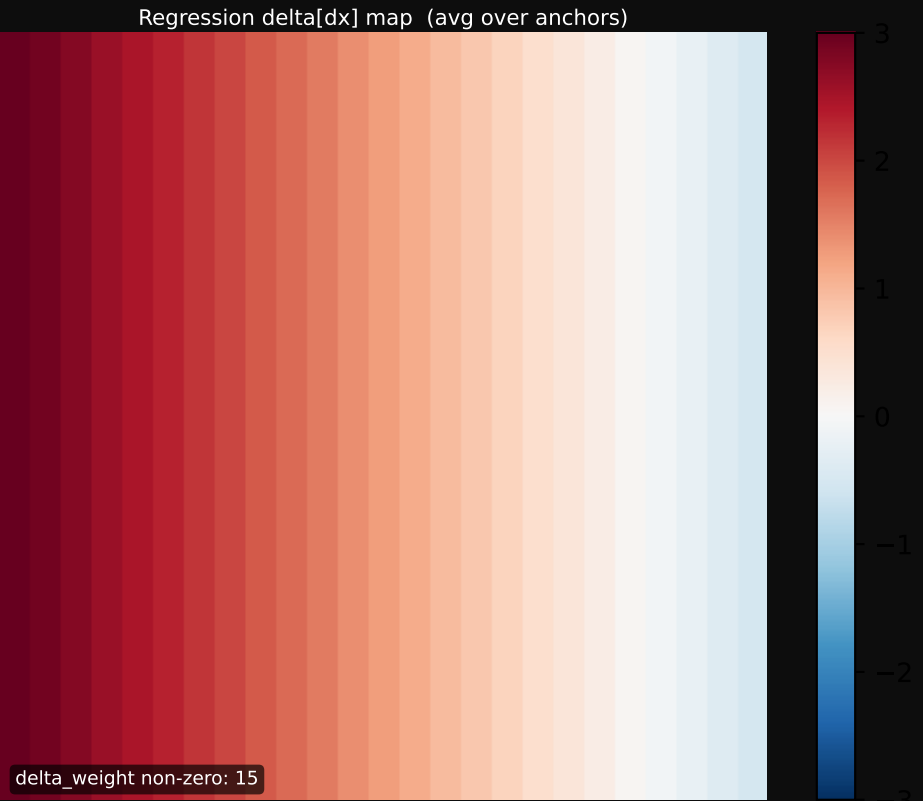
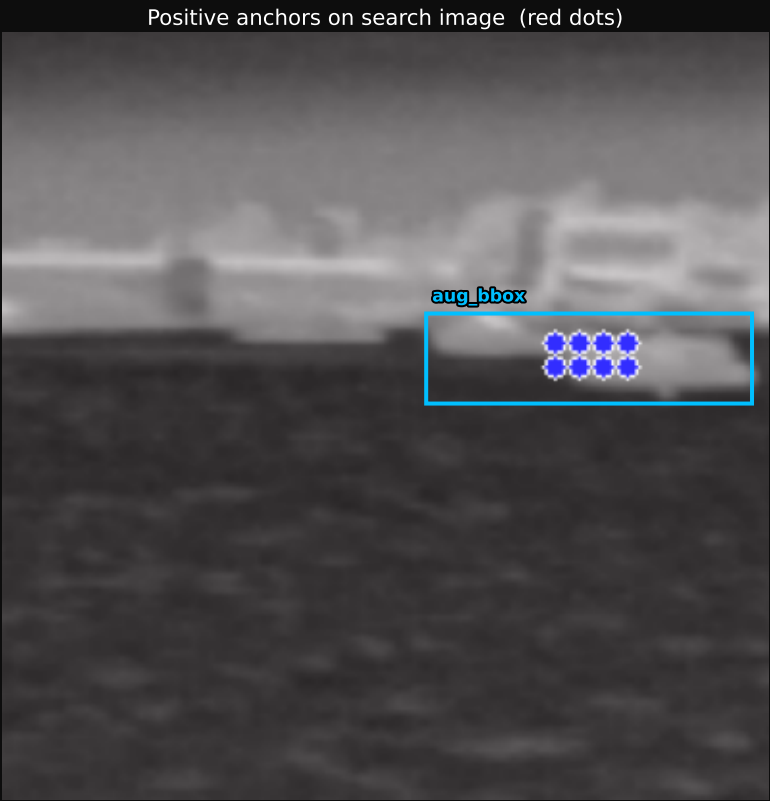
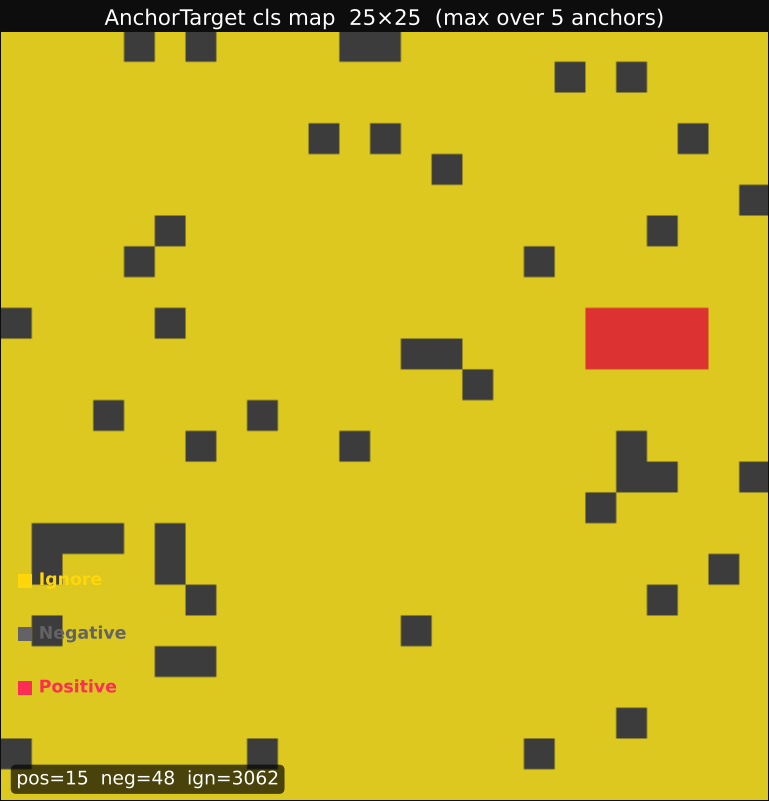


```
— Augmentation config —
TEMPLATE:
SHIFT=4px
SCALE=5%
BLUR=0.0
FLIP=0.0

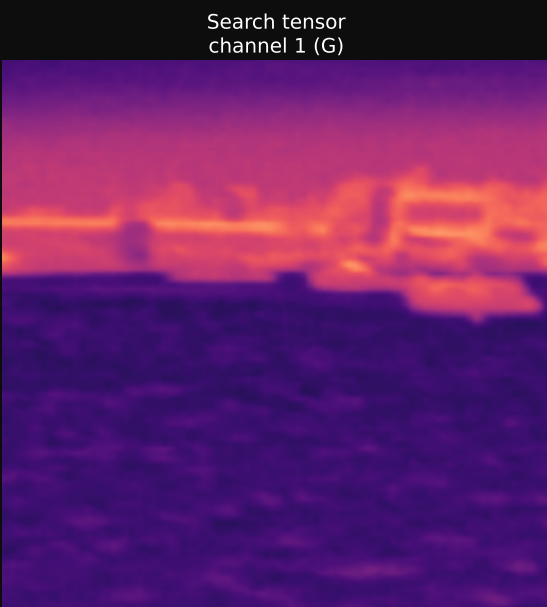
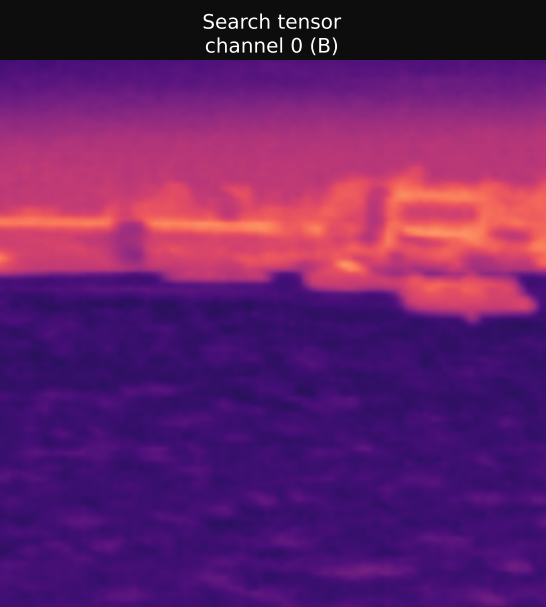
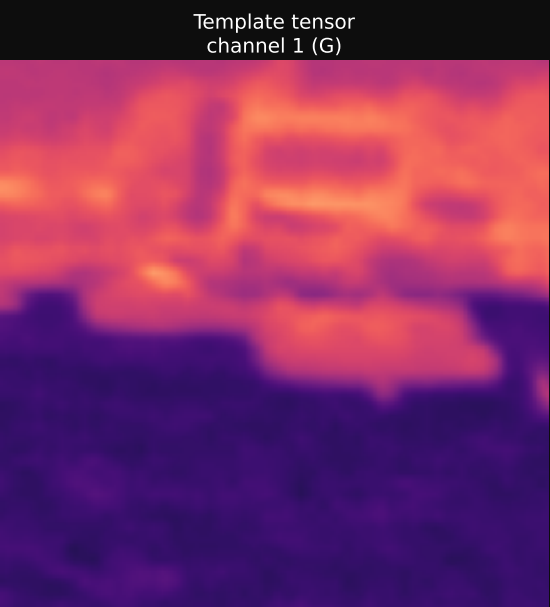
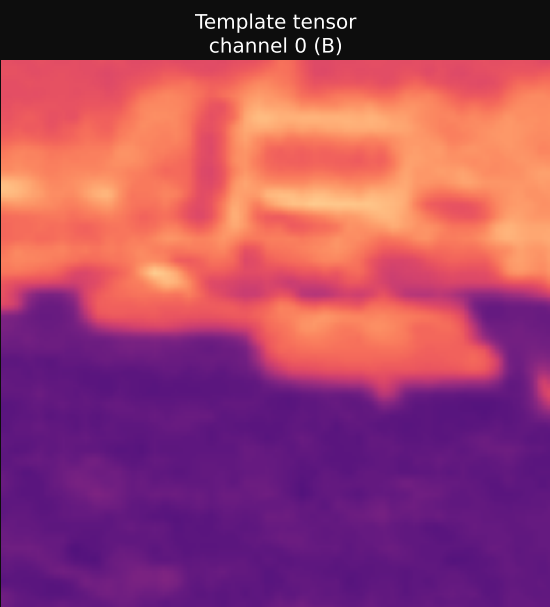
SEARCH:
SHIFT=64px
SCALE=18%
BLUR=0.0
FLIP=0.0

— aug_bbox (to anchor target) —
x1=140.53 y1=92.83
x2=248.87 y2=122.76
w=108.34
h=29.94
cx=194.70
cy=107.79
```

STEP 3 – ANCHOR TARGET



STEP 4 – MODEL TENSORS



```
— Template tensor (→ model) —
shape : (3, 127, 127)
dtype : float32
min : 38.908
max : 228.315
mean : 112.630
std : 53.642
range : 189.608

— Search tensor (→ model) —
shape : (3, 255, 255)
dtype : float32
min : 38.908
max : 203.800
mean : 84.831
std : 41.031
range : 165.201

— Note —
Raw 0-255 BGR float32
NO /255, NO mean-sub
(matches training-time
PySOT convention)
```

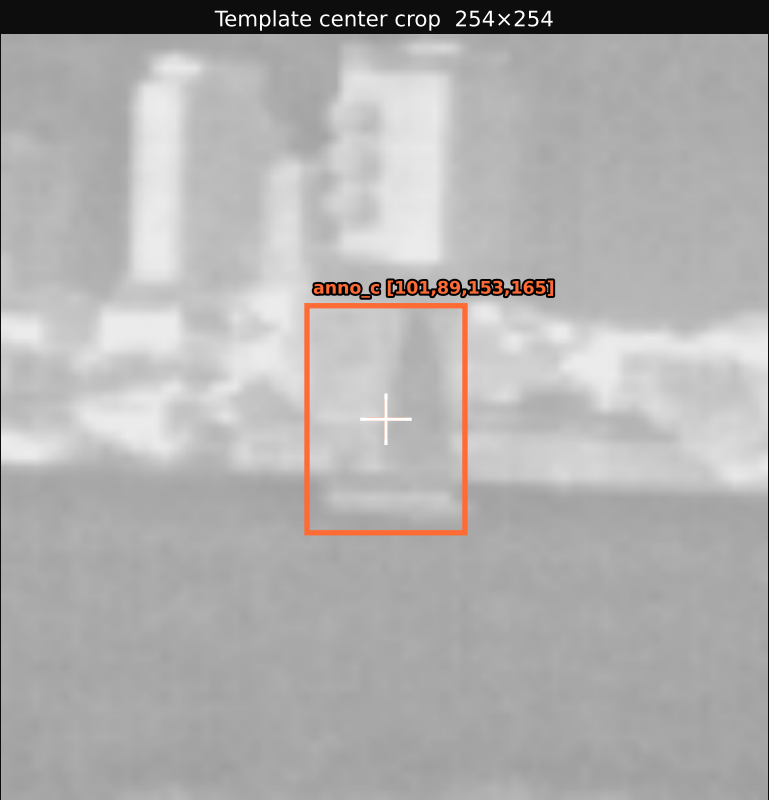
STEP 0 – RAW INPUT



```
— TEMPLATE ANNOTATION —
Format : [x1,y1,x2,y2]
Values : [408.0, 249.0, 416.0, 272.0]
Size : 16.0 × 23.0 px
Center : (408.0, 260.5)
AR : 0.78
Image : 640×512

— SEARCH ANNOTATION —
Values : [408.0, 249.0, 416.0, 272.0]
Size : 16.0 × 23.0 px
Center : (408.0, 260.5)
Image : 640×512
```

STEP 1 – CENTER CROP

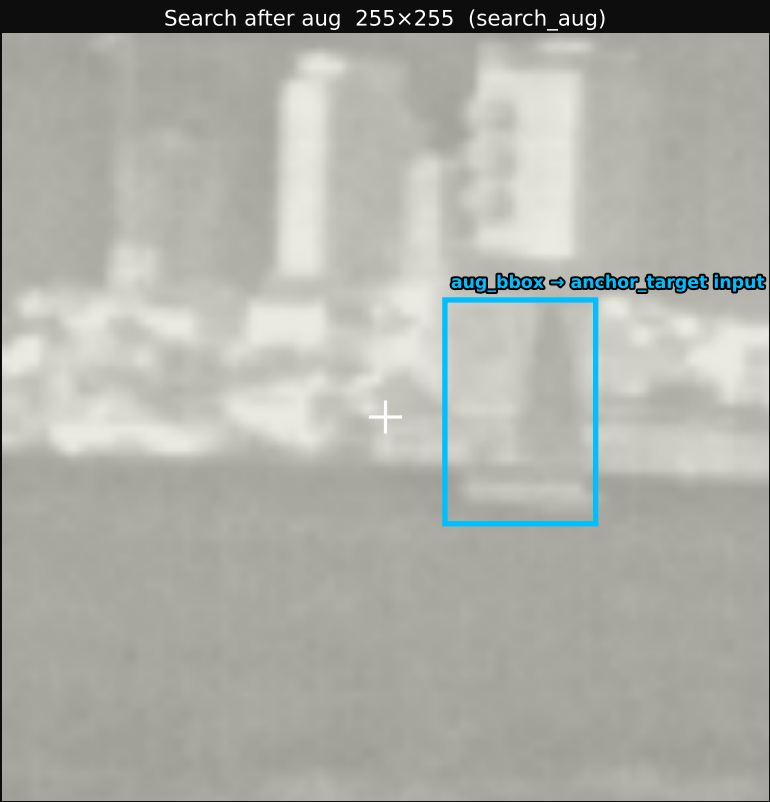
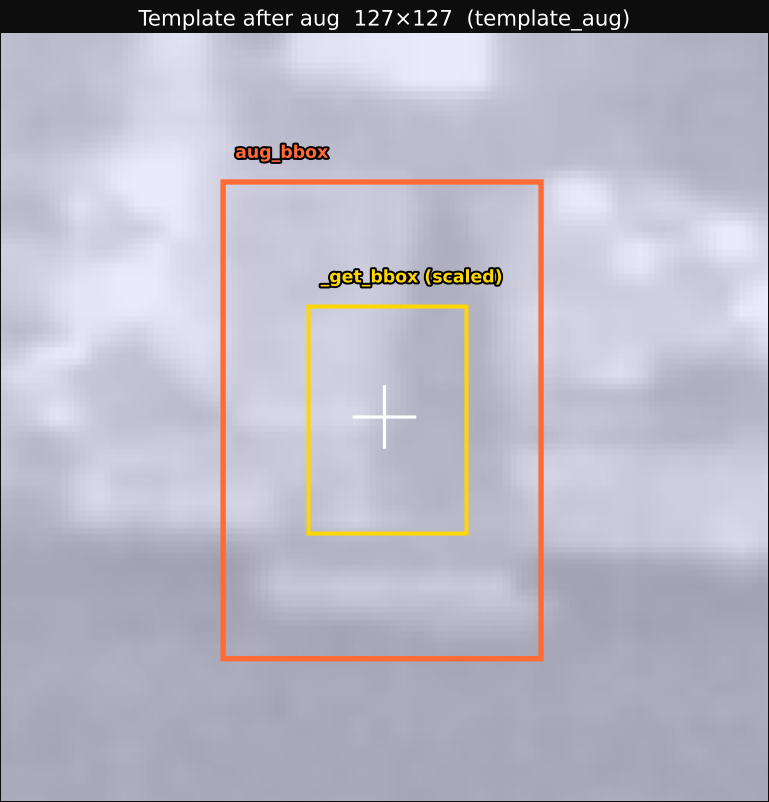


```
— get_center_crop —
Output size (tmpl): 254×254
Output size (srch): 510×510
White cross = image centre
Orange box = new annotation

— get_bbox_output —
TEMPLATE (corner):
[100.8, 89.4, 153.2, 164.6]
cx=127.0 cy=127.0
w=52.3 h=75.2

SEARCH (corner):
[228.8, 217.4, 281.2, 292.6]
cx=255.0 cy=255.0
w=52.3 h=75.2
```

STEP 2 – AUGMENTATION

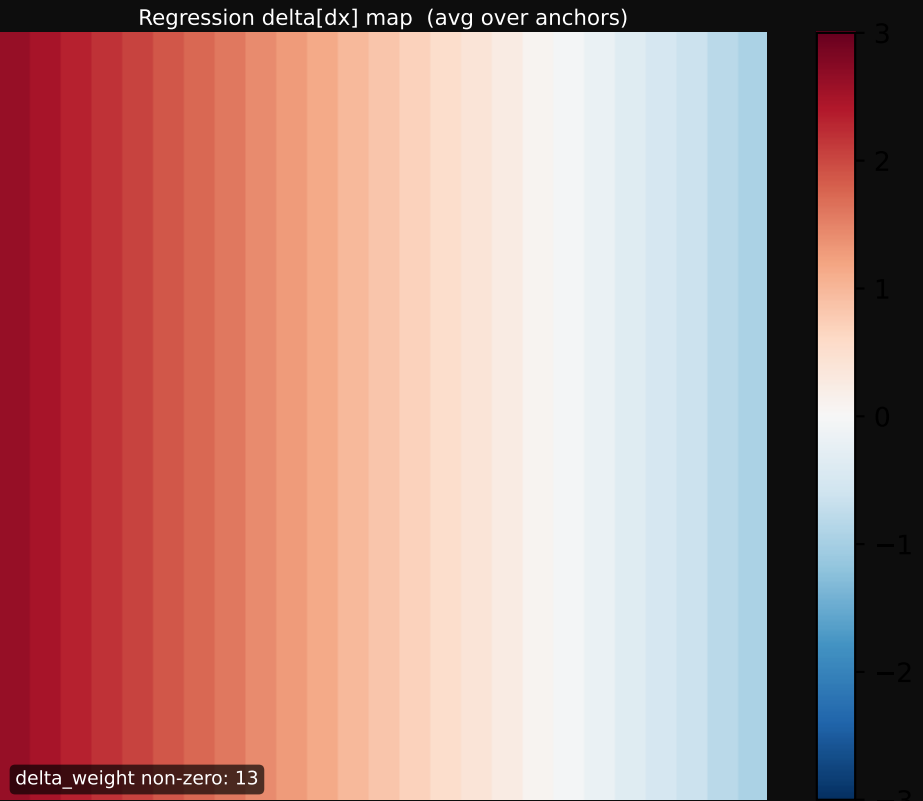
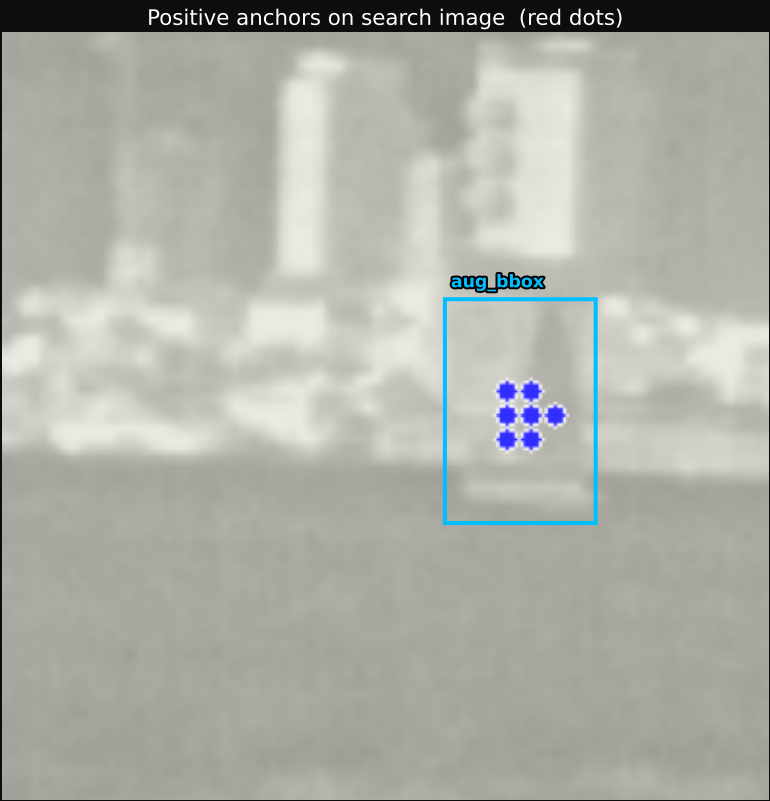


```
— Augmentation config —
TEMPLATE:
SHIFT=4px
SCALE=5%
BLUR=0.0
FLIP=0.0

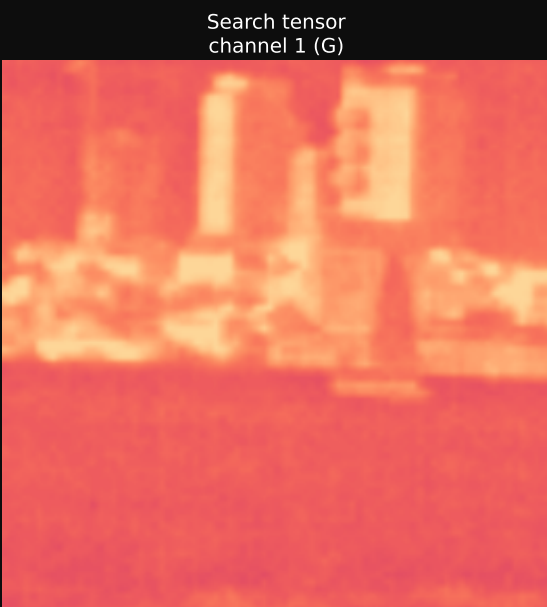
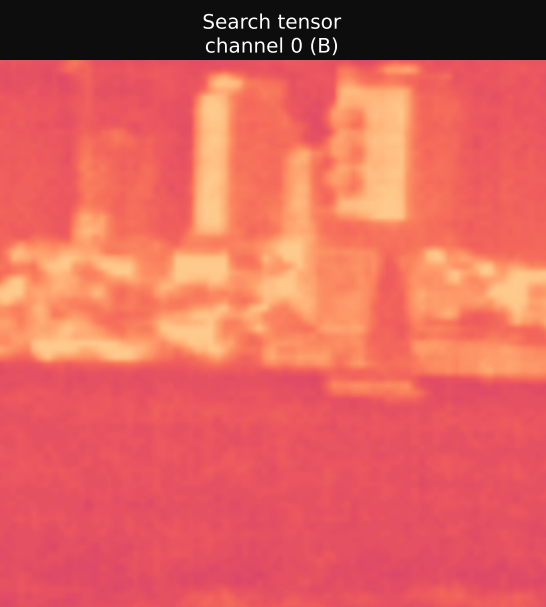
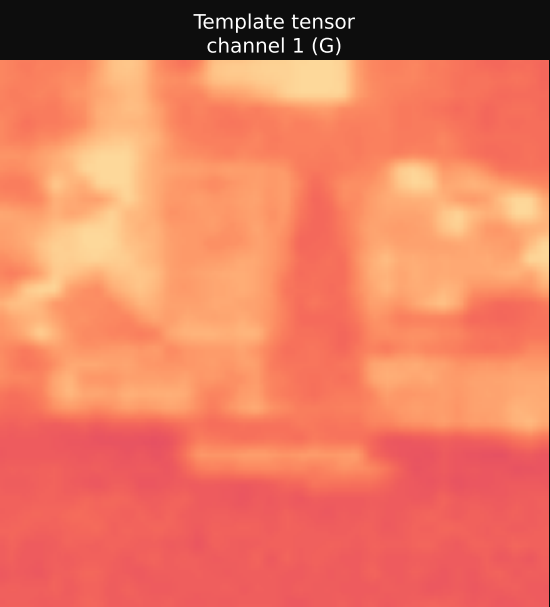
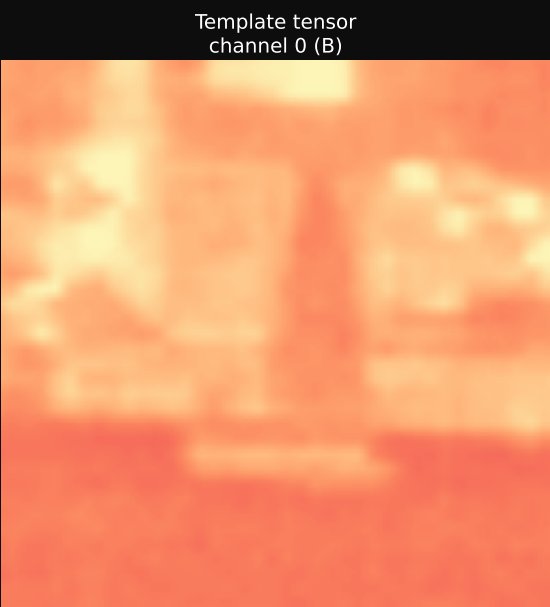
SEARCH:
SHIFT=64px
SCALE=18%
BLUR=0.0
FLIP=0.0

— aug_bbox (to anchor target) —
x1=146.76 y1=88.07
x2=196.88 y2=162.49
w=50.13
h=74.42
cx=171.82
cy=125.28
```

STEP 3 – ANCHOR TARGET



STEP 4 – MODEL TENSORS



```
— Template tensor (→ model) —
shape : (3, 127, 127)
dtype : float32
min : 160.225
max : 250.244
mean : 195.594
std : 19.685
range : 90.019

— Search tensor (→ model) —
shape : (3, 255, 255)
dtype : float32
min : 149.686
max : 235.191
mean : 180.698
std : 19.657
range : 85.506

— Note —
Raw 0-255 BGR float32
NO /255, NO mean-sub
(matches training-time
PySOT convention)
```

● GT annotation (raw frame)

● Annotation after center-crop / aug_bbox

● _get_bbox output (scaled to aug input)

● aug_bbox -> anchor_target input

● Image centre crosshair

● Positive anchor

STEP 0 - RAW INPUT



```
— TEMPLATE ANNOTATION —
Format : [x1,y1,x2,y2]
Values : [1051.0, 374.0, 1313.0, 960.0]
Size : 262.0 x 586.0 px
Center : (1192.0, 667.0)
AR : 0.45
Image : 1920x1080

— SEARCH ANNOTATION —
Values : [781.0, 524.0, 1061.0, 1080.0]
Size : 280.0 x 556.0 px
Center : (921.0, 802.0)
Image : 1920x1080
```

STEP 1 - CENTER CROP



```
— get_center_crop —
Output size (tmpl): 254x254
Output size (srch): 510x510
White cross = image centre
Orange box = new annotation

— get_bbox_output —
TEMPLATE (corner):
[107.0, 82.3, 147.0, 171.7]
cx=127.0 cy=127.0
w=40.0 h=89.4

SEARCH (corner):
[233.4, 212.2, 276.6, 297.8]
cx=255.0 cy=255.0
w=43.1 h=85.6
```

STEP 2 - AUGMENTATION

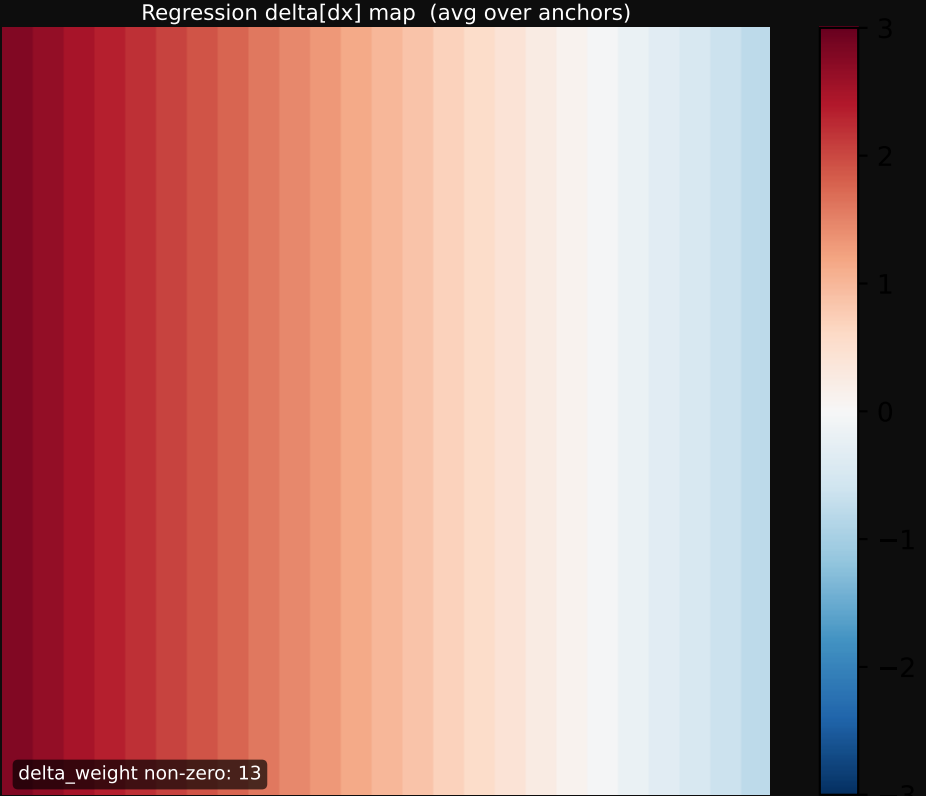
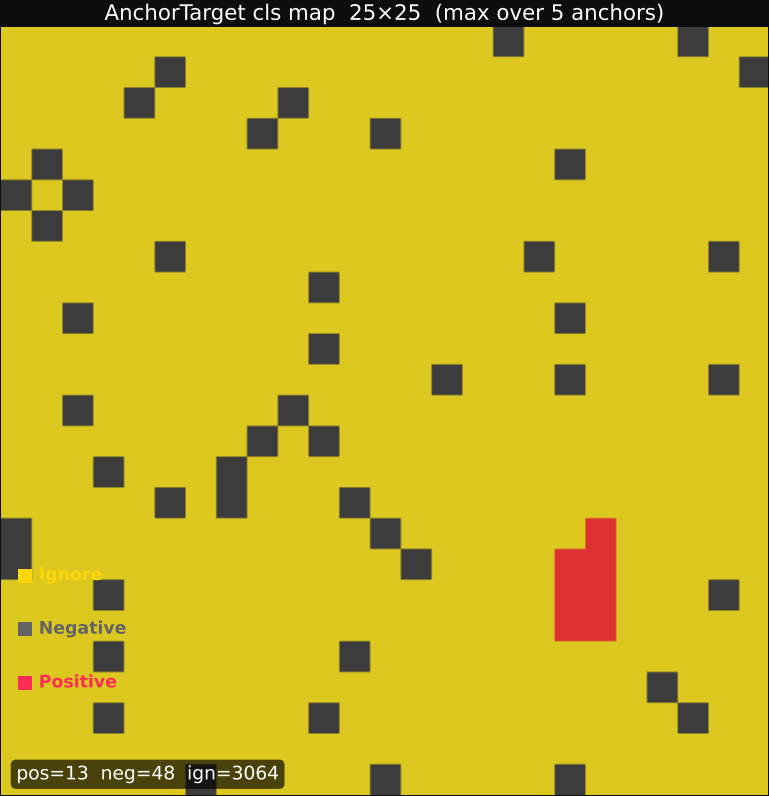


```
— Augmentation config —
TEMPLATE:
SHIFT=4px
SCALE=5%
BLUR=0.0
FLIP=0.0

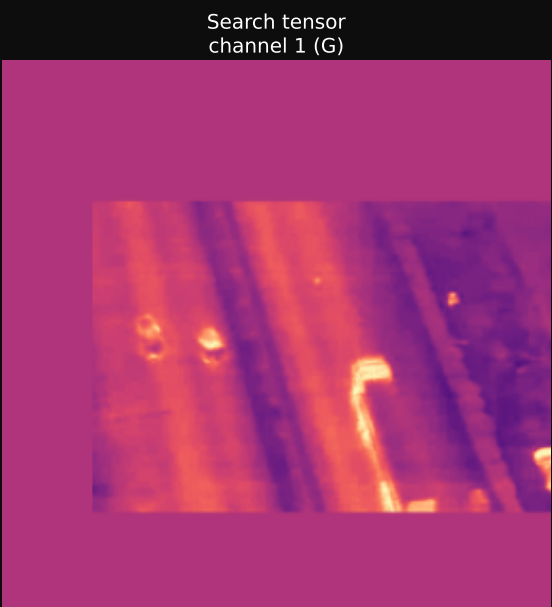
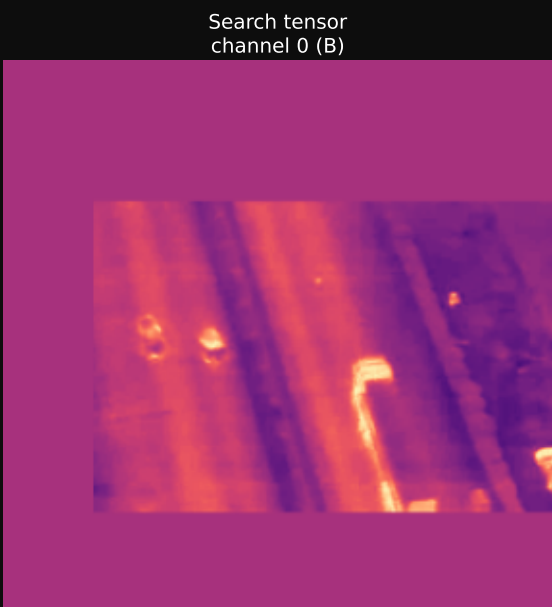
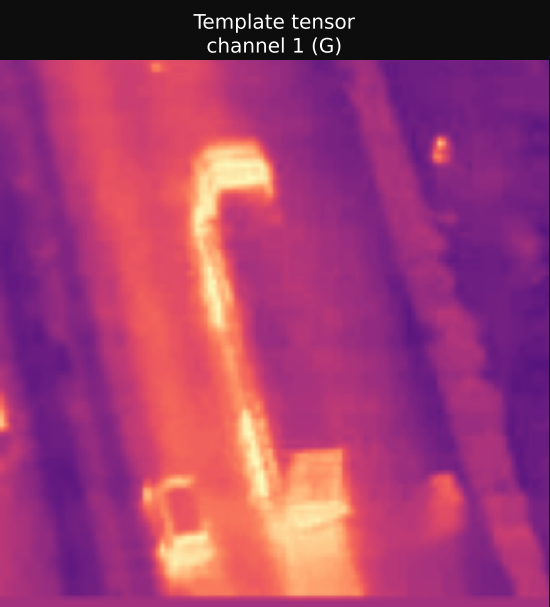
SEARCH:
SHIFT=64px
SCALE=10%
BLUR=0.0
FLIP=0.0

— aug_bbox (to anchor target) —
x1=159.35 y1=135.70
x2=201.64 y2=209.96
w=42.10
h=74.26
cx=180.59
cy=172.83
```

STEP 3 - ANCHOR TARGET



STEP 4 - MODEL TENSORS



```
— Template tensor (→ model) —
shape : (3, 127, 127)
dtype : float32
min : 52.400
max : 256.375
mean : 120.625
std : 39.217
range : 203.976

— Search tensor (→ model) —
shape : (3, 255, 255)
dtype : float32
min : 59.227
max : 255.834
mean : 123.132
std : 20.880
range : 196.607

— Note —
Raw 0-255 BGR float32
NO /255, NO mean-sub
(matches training-time
PySOT convention)
```

STEP 0 - RAW INPUT



```
— TEMPLATE ANNOTATION —
Format : [x1,y1,x2,y2]
Values : [741.0, 386.0, 786.0, 452.0]
Size : 45.0 x 66.0 px
Center : (763.5, 419.0)
AR : 0.68
Image : 1920x1080

— SEARCH ANNOTATION —
Values : [741.0, 367.0, 786.0, 433.0]
Size : 45.0 x 66.0 px
Center : (763.5, 400.0)
Image : 1920x1080
```

STEP 1 - CENTER CROP



```
— get_center_crop —
Output size (tmpl): 254x254
Output size (srch): 510x510
White cross = image centre
Orange box = new annotation

— get_bbox_output —
TEMPLATE (corner):
[101.1, 89.1, 152.9, 164.9]
cx=127.0 cy=127.0
w=51.7 h=75.9

SEARCH (corner):
[229.1, 217.1, 280.9, 292.9]
cx=255.0 cy=255.0
w=51.7 h=75.9
```

STEP 2 - AUGMENTATION

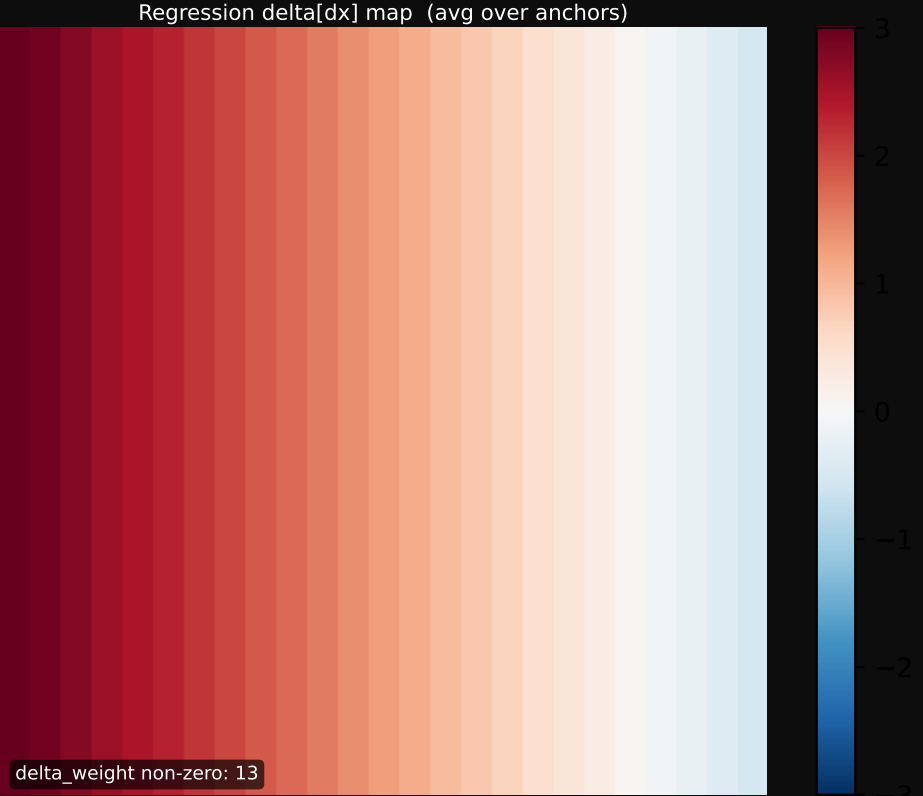
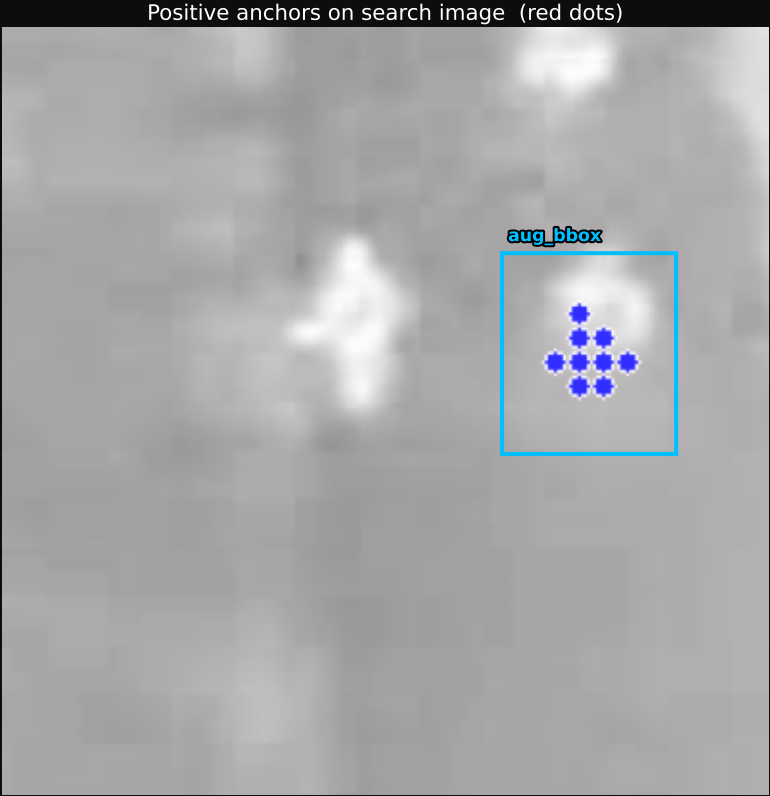
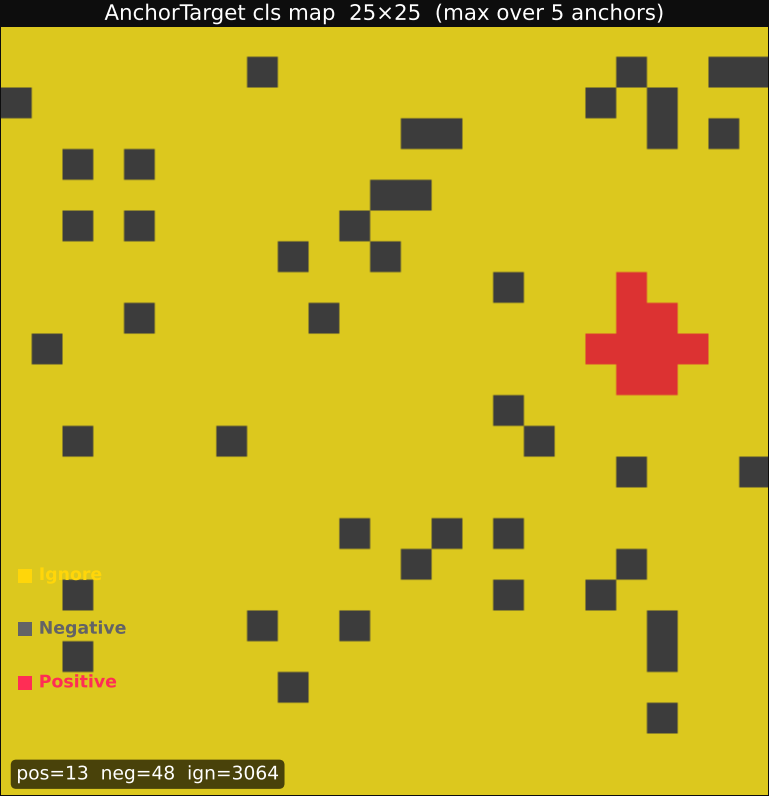


```
— Augmentation config —
TEMPLATE:
SHIFT=4px
SCALE=5%
BLUR=0.0
FLIP=0.0

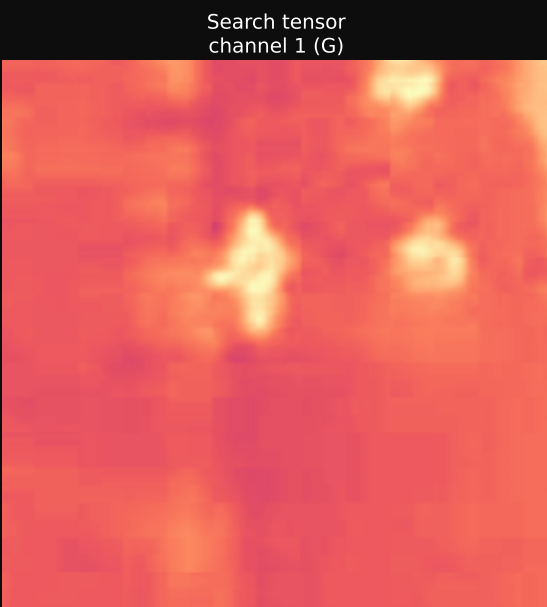
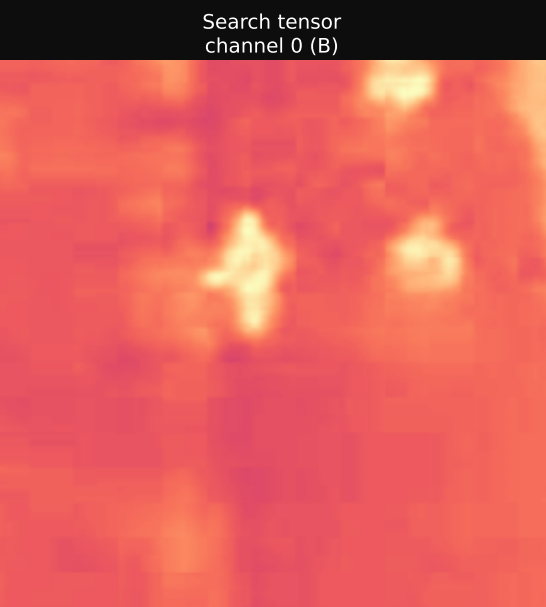
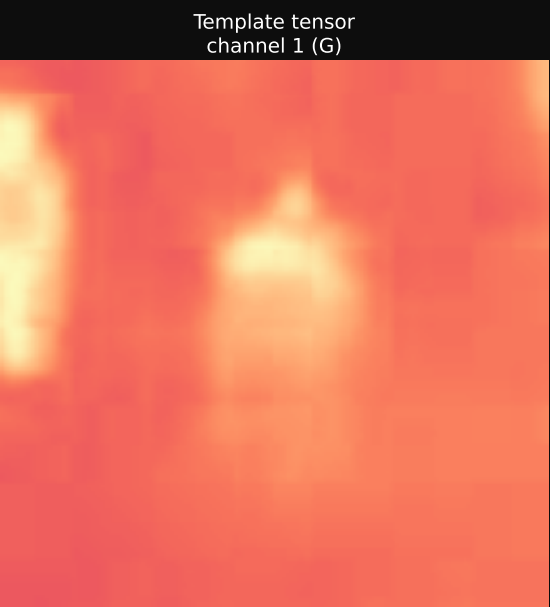
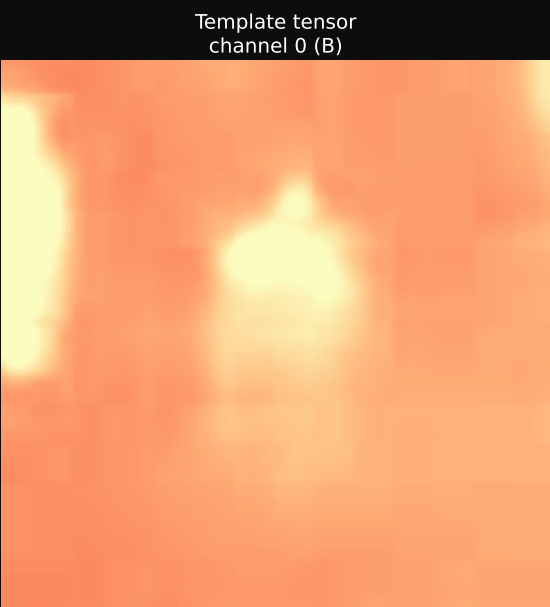
SEARCH:
SHIFT=64px
SCALE=18%
BLUR=0.0
FLIP=0.0

— aug_bbox (to anchor target) —
x1=165.76 y1=74.41
x2=223.64 y2=141.18
w=57.88
h=66.77
cx=194.70
cy=107.79
```

STEP 3 - ANCHOR TARGET



STEP 4 - MODEL TENSORS



```
— Template tensor (→ model) —
shape : (3, 127, 127)
dtype : float32
min : 163.908
max : 279.516
mean : 194.885
std : 21.439
range : 115.608

— Search tensor (→ model) —
shape : (3, 255, 255)
dtype : float32
min : 144.598
max : 256.800
mean : 174.174
std : 15.437
range : 112.201

— Note —
Raw 0-255 BGR float32
NO /255, NO mean-sub
(matches training-time
PySOT convention)
```

STEP 0 - RAW INPUT



```
— TEMPLATE ANNOTATION —
Format : [x1,y1,x2,y2]
Values : [1461.0, 776.0, 1609.0, 1037.0]
Size : 148.0 x 261.0 px
Center : (1535.0, 906.5)
AR : 0.57
Image : 1920x1080

— SEARCH ANNOTATION —
Values : [1151.0, 264.0, 1290.0, 455.0]
Size : 145.0 x 191.0 px
Center : (1223.5, 359.5)
Image : 1920x1080
```

STEP 1 - CENTER CROP



```
— get_center_crop —
Output size (tmpl): 254x254
Output size (srch): 510x510
White cross = image centre
Orange box = new annotation

— get_bbox_output —
TEMPLATE (corner):
[103.8, 86.1, 150.2, 167.9]
cx=127.0 cy=127.0
w=46.4 h=81.8

SEARCH (corner):
[227.5, 218.8, 282.5, 291.2]
cx=255.0 cy=255.0
w=54.9 h=72.4
```

STEP 2 - AUGMENTATION

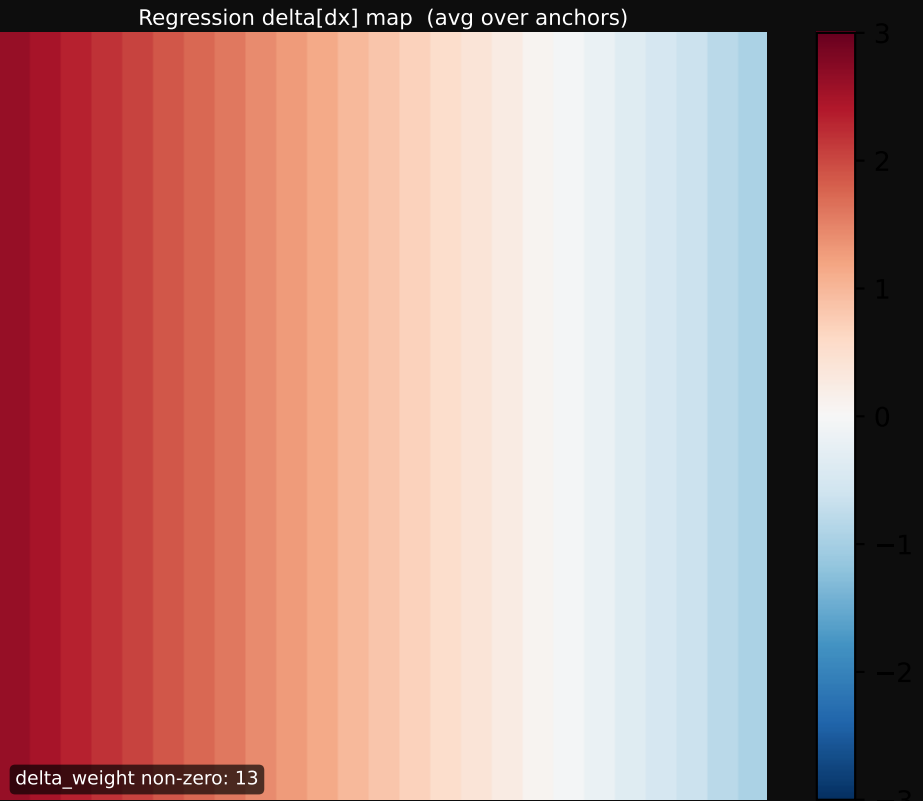


```
— Augmentation config —
TEMPLATE:
SHIFT=4px
SCALE=5%
BLUR=0.0
FLIP=0.0

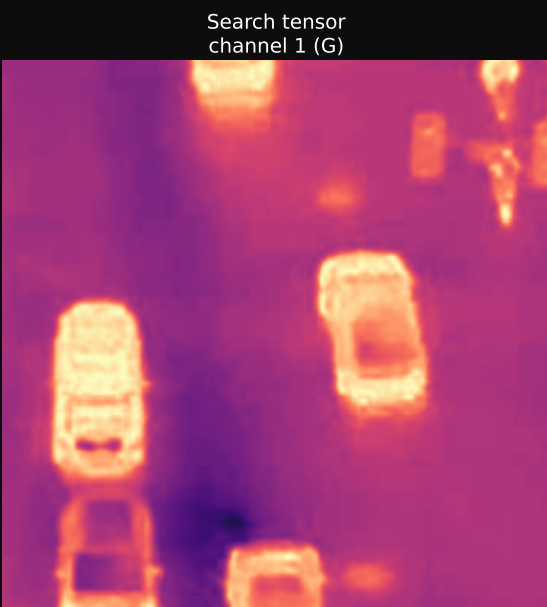
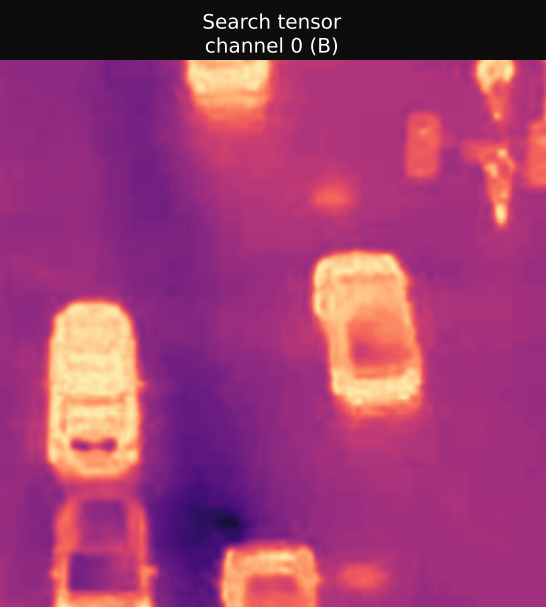
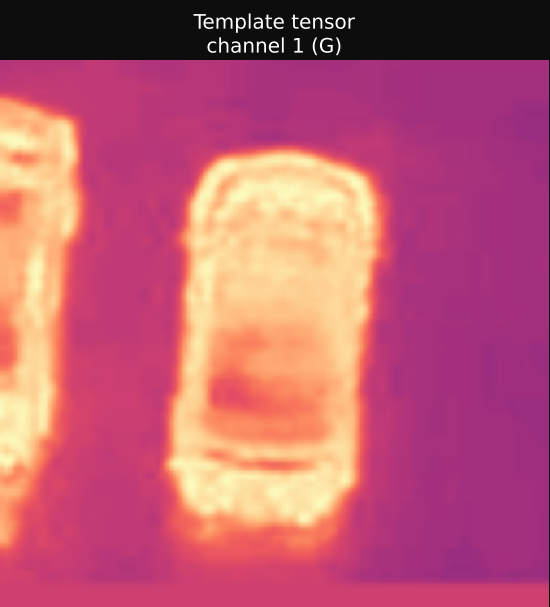
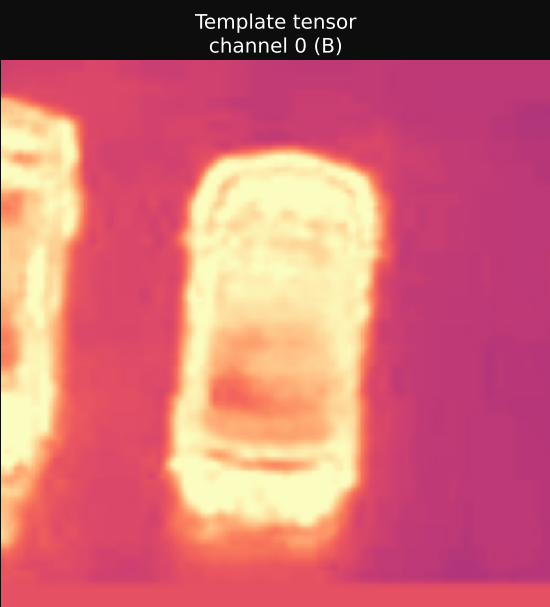
SEARCH:
SHIFT=64px
SCALE=18%
BLUR=0.0
FLIP=0.0

— aug_bbox (to anchor target) —
x1=145.80 y1=89.47
x2=198.14 y2=161.09
w=52.64
h=71.61
cx=171.92
cy=125.28
```

STEP 3 - ANCHOR TARGET



STEP 4 - MODEL TENSORS



```
— Template tensor (← model) —
shape : (3, 127, 127)
dtype : float32
min : 188.225
max : 270.244
mean : 163.953
std : 45.298
range : 162.019

— Search tensor (← model) —
shape : (3, 255, 255)
dtype : float32
min : 26.686
max : 255.191
mean : 132.492
std : 39.878
range : 228.506

— Note —
Raw 0-255 BGR float32
NO /255, NO mean-sub
(matches training-time
PySOT convention)
```